राष्ट्रीय भू-सूचना विज्ञान एवं प्रौद्योगिकी संस्थान
भारतीय सर्वेक्षण विभाग
विज्ञान और प्रौद्योगिकी विभाग

National Institute of Geo-Informatics Science & Technology
Survey of India
Department of Science & Technology

STANDARD OPERATING PROCEDURE

for

Field Operation of

RTK Method using

(Trimble R980 GNSS & Trimble Access)

Course Name				
Training Module Name	**RTK Method of GNSS Observations**			
Description	**RTK Method using Trimble R980 & Trimble Access**			
Created By	MANUJ KUMAR, Surveyor, Faculty of Geodesy, NIGST			
Date Created	24.12.2025			
Maintained By	Faculty of Geodesy, NIGST			
Version Number	Modified By	Modifications Made	Date Modified	Status
1	MANUJ KUMAR , Surveyor			

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1. Introduction

This Standard Operating Procedure (SOP) provides step-by-step instructions for conducting RTK surveys using the Trimble R980 GNSS receiver and Trimble Access software. Prepared by NIGST, this SOP aims to guide users through essential functions, from instrument setup to data export, ensuring accurate data collection and efficient workflow. As users gain familiarity with the equipment, they are encouraged to explore advanced features to further enhance survey productivity.

2. Objective

The main objectives of conducting an **RTK GNSS survey using the Trimble R980 receiver** are:

- **Achieve Centimetre-Level Accuracy**
  To obtain **high-precision (1–2 cm)** real-time position coordinates using multi-frequency and multi-constellation GNSS tracking.
- **Real-Time Data Collection**
  To collect accurate survey data **instantly in the field**, reducing post-processing time and improving productivity.
- **Establish Control and Reference Points**
  To create or densify **survey control points** for mapping, construction, and engineering projects.
- **Efficient Field Operations**
  To perform fast and reliable surveys with the **IMU-based tilt compensation** of Trimble R980, minimizing pole levelling time.
- **Reliable Performance in Challenging Environments**
  To maintain accuracy in areas with **multipath, obstructions, or interference** using Trimble ProPoint™ GNSS technology.
- **Support Multiple Correction Sources**
  To utilize corrections from **local base stations & radio** for flexible survey setups.
- **Improve Project Accuracy and Quality Control**
  To ensure consistent, repeatable results and enable **real-time quality checks** during data capture.

3. Note on RTK

RTK (Real-Time Kinematic) GNSS is a high-precision satellite positioning technique used to achieve **centimetre-level accuracy** in real time.

RTK works by using **two GNSS receivers**:

- **Base Station** – placed on a known, fixed point
- **Rover** – moves to measure unknown points

The base station calculates GNSS errors (satellite clock, orbit, atmospheric delays) and sends **correction data** to the rover via **radio, GSM, or internet (NTRIP)**.

The rover applies these corrections instantly and resolves **carrier-phase ambiguities**, allowing very precise positioning.

Key Features

- Accuracy: **1–2 cm (horizontal)**, **2–3 cm (vertical)**
- Real-time results
- Uses carrier-phase measurements
- Requires continuous communication between base and rover

Applications

- Land surveying
- Construction and machine control
- GIS data collection
- Precision agriculture
- Deformation monitoring

Limitations

- Limited range (typically **10–30 km** from base)
- Signal obstruction affects performance
- Needs reliable data link

RTK GNSS is widely used where **high accuracy and real-time positioning** are essential.

4. Hardware & Software

4.1 Hardware

➢ **Trimble R980 GNSS Receiver**

a) Multi-frequency, multi-constellation GNSS receiver (GPS, GLONASS, Galileo, BeiDou, QZSS, NavIC-ready)
b) Used as **Base or Rover**
c) IMU-based tilt compensation

➢ **GNSS Antenna (Integrated)**

a) High-precision antenna built into R980 for accurate phase tracking

➢ **Survey Controller / Data Collector**

a) Trimble TSC7 / TSC5 / TCU5
b) Runs field software and controls the receiver

➢ **Communication Devices**

a) **UHF Radio** (internal or external) for Base–Rover link
b) **GSM / LTE modem** for Network RTK (NTRIP/VRS)

- **Tripod & Tribrach**

a) For stable base station setup

- **Survey Pole**

a) For rover observations

- **Batteries & Chargers**

a) Internal and external power supply for long field operations

- **Cables & Accessories**

a) USB, antenna cables, radio cables (if external radio used)

4.2 Software

- **Trimble Access (Field Software)**

  Used on controller for:

  - RTK survey setup
  - Base and rover configuration
  - Point measurement and stakeout
  - Coordinate system management

- **Firmware (Receiver Internal Software)**

- Controls GNSS tracking, radio, IMU, and communications
- Must be compatible with Trimble Access and TBC

5. Pre requisites

- **GNSS Equipment**

- RTK-capable GNSS receivers (Base and Rover) .
- Multi-frequency, multi-constellation support for reliable fixes.

- **Known Reference Point**

- Base station coordinates must be known accurately

- **Communication Link**

- Reliable data link between base and rover:
    - **UHF radio**, or
    - **GSM / Internet** with good mobile network coverage

- **Clear Satellite Visibility**

- Open sky with minimal obstructions (trees, buildings)
- Good satellite geometry (**low PDOP**)

- **Software & Licensing**

- Field software (e.g., **Trimble Access**)
- Compatible **receiver firmware**

- **Stable Base Setup**

- Firm tripod and tribrach
- Correct antenna height measurement
- Base station should not move during survey

- **Power Supply**

- Fully charged batteries for base, rover, controller, and radios

- **Survey Planning**

- Check satellite availability and DOP values
- Select appropriate observation time

- **Operator Knowledge**

- Understanding of RTK principles
- Correct handling of equipment and settings

6. Roles & Responsibilities

During an **RTK GNSS survey**, clear roles and responsibilities are essential to ensure **accuracy, efficiency, and data quality**.

A field Observer shall:

- Plan the RTK survey methodology
- Select base station location or RTK network
- Ensure correct coordinate system and datum
- Set up the base station on a known control point
- Accurately measure and record antenna height
- Configure base settings (radio/GSM, corrections)
- Monitor power supply and ensure base stability

- Ensure continuous transmission of corrections
- Configure rover receiver and controller
- Check FIX status, PDOP, and satellite tracking
- Measure points accurately using proper pole height
- Avoid obstructions and multipath areas
- Manage job files and point naming
- Apply correct feature codes and attributes
- Perform stakeout and measurements as per design
- Ensure data is saved and backed up properly
- Perform check shots on known points
- Monitor accuracy indicators (HRMS, VRMS)
- Maintain survey logs and field notes
- Ensure safe working conditions on site
- Manage equipment transport and protection
- Coordinate batteries, chargers, and accessories
- Import RTK data into office software (e.g., TBC, CAD etc.) & secure it.
- Generate reports, drawings, and deliverables

7. Specifications of Trimble R980 GNSS Receiver

The Trimble R980 GNSS system is a premier solution for land surveyors, offering exceptional accuracy and efficiency. Featuring the Trimble ProPoint® GNSS engine and Trimble Inertial Platform™ (TIP™) IMU-based tilt compensation, users can collect precise data in challenging environments without leveling the pole. Below are the specifications of Trimble R980 GNSS Receiver:

- **Performance specifications**
  - **GNSS TECHNOLOGY**
    1) Constellation agnostic, flexible signal tracking, improved positioning in challenging environments and inertial measurement integration with Trimble ProPoint® GNSS technology.
    2) Increased measurement and stakeout productivity and traceability with Trimble Inertial Platform™ (TIP™) technology IMU-based tilt compensation.
    3) Dual Trimble Maxwell™ 7 Custom GNSS chips with 672 channels.
    4) Trimble EVEREST™ Plus multipath signal rejection.
    5) Trimble IonoGuard™ technology for mitigation of ionospheric GNSS signal disruptions.
    6) Trimble CenterPoint® RTX correction service is activated and ready to use for the initial 12 months.
    7) Spectrum Analyzer to troubleshoot GNSS jamming.
    8) Digital Signal Processor (DSP) techniques to detect and recover from spoofed GNSS signals.

- **SATELLITE TRACKING**
  GPS: L1C, L1C/A, L2C, L2E, L5
  GLONASS: L1C/A, L1P, L2C/A, L2P, L3
  SBAS (WAAS, EGNOS, GAGAN, MSAS, SDCM): L1C/A, L5
  Galileo: E1, E5A, E5B, E5 AltBOC, E62
  BeiDou: B1I, B1C, B2I, B2A, B2B, B3I
  QZSS: L1C/A, L1S, L1C, L2C, L5, L6
  NavIC (IRNSS): L5
  L-band: Trimble RTX® Corrections

➢ Positioning performance

- **STATIC GNSS SURVEYING**

	Horizontal	Vertical
High-Precision Static	3 mm + 0.1 ppm RMS	3.5 mm + 0.4 ppm RMS
Static and Fast Static	3 mm + 0.5 ppm RMS	5 mm + 0.5 ppm RMS

- **REAL TIME KINEMATIC SURVEYING**

	Horizontal	Vertical
Single Baseline < 30 km	8 mm + 1 ppm RMS	15 mm + 1 ppm RMS
Network RTK	8 mm + 0.5 ppm RMS	15 mm + 0.5 ppm RMS
RTK start-up time for specified precisions	2 to 8 seconds	

- **TRIMBLE INERTIAL PLATFORM (TIP) TECHNOLOGY**

	Horizontal
TIP Compensated Surveying	RTK + 3mm + 0.15mm/°tilt (up to 40°) RMS
	RTX + 3mm + 0.15mm/°tilt (up to 40°) RMS

- **TRIMBLE RTX CORRECTION SERVICES**

CenterPoint RTX	Horizontal	2 cm RMS
	Vertical	3 cm RMS
	Convergence time for specified precisions in Trimble RTX Fast regions	< 1 min
	Convergence time for specified precisions in non Trimble RTX Fast regions	< 3 min
	Quick Start convergence time for specified precisions	< 1 min

- **TRIMBLE xFill**

	Horizontal	Vertical
TRIMBLE xFill	RTK + 10 mm/minute RMS	RTK + 20 mm/minute RMS

➢ Hardware

- **PHYSICAL**

Dimensions (W×H)	11.9 cm x 13.6 cm (4.6 in x 5.4 in)
Weight	0.84 kg (1.85 lb) receiver only, no radio model
	0.86 kg (1.90 lb) receiver only, radio model
	1.13 kg (2.49 lb) total weight including internal battery and UHF antenna
	3.96 kg (8.73 lb) items above plus range pole, Trimble TSC7 data collector and bracket

- **TEMPERATURE**

Operating	-40 °C to +65 °C (-40 °F to +149 °F)
Storage	-40 °C to +80 °C (-40 °F to +176 °F)

- **OPERATING TIMES ON INTERNAL BATTERY**

Rover	450 or 900 MHz receive	5.5–6.3 hours
	Cellular receive (Internal or Controller via Bluetooth)	7.0 hours
Base station	450 MHz transmit	3.7–5.8 hours (1.0 W transmit available only where legally permitted)
	900 MHz transmit (1.0 W)	6.0 hours (900 MHz transmit available only where legally permitted)
	Cellular transmit	7.0 hours

- **Communications and data storage**

Radio modem	Transmit power (Internal Radio)	0.1 W, 0.5 W, 1.0 W (1.0 W available only where legally permitted)
	Range	3-5 km typical with Internal Radio, 10 km optimal with External Radio
Cellular	Fully integrated, fully-sealed LTE compliant module with 2G/3G fallback	
Bluetooth	Fully-integrated, fully-sealed 2.4 GHz Bluetooth module	
Wi-Fi	Fully-integrated, fully-sealed 2.4 GHz Wi-Fi module	
Positioning rates	1 Hz, 2 Hz, 5 Hz, 10 Hz, and 20 Hz	
I/O ports	Serial, USB, TCP/IP, IBSS/NTRIP, Bluetooth	
Data storage	9 GB internal memory	

	**1** SMA connection: UHF antenna
	2 Quick release socket
	3 Lemo port 1: Serial connection
	4 Lemo port 2: USB connection

Receiver ports

Icon	Name	Connections
IOIOI	Port 1	Device, computer, external radio, power in
	Port 2	Device, computer, USB flash memory stick, power in, power out
	RADIO	Radio communications antenna

<u>CONTROLLER</u>

TRIMBLE R980 RTK-EXTERNAL RADIO FIELD SETUP

Trimble R980 Internal Radio Field Setup

8. Setting up the GNSS Equipment

8.1 Setting up Base Station

- Fix tripod firmly over known control point
- Mount Trimble R980 on tripod using tribrach & extension lever.
- Center using optical plummet
- Level the tripod accurately
- Attach UHF Radio Antenna to Base Receiver
- If using External Radio then attach it to R980 receiver and with External Broadcasting Antenna.
- Insert internal/external batteries in Base Receiver & also provide External Power Supply to External Radio (If using).
- Measure height of Base Receiver up to Extension Lever using Measuring Tape.
- Switch On Base Receiver and External Radio (If using).

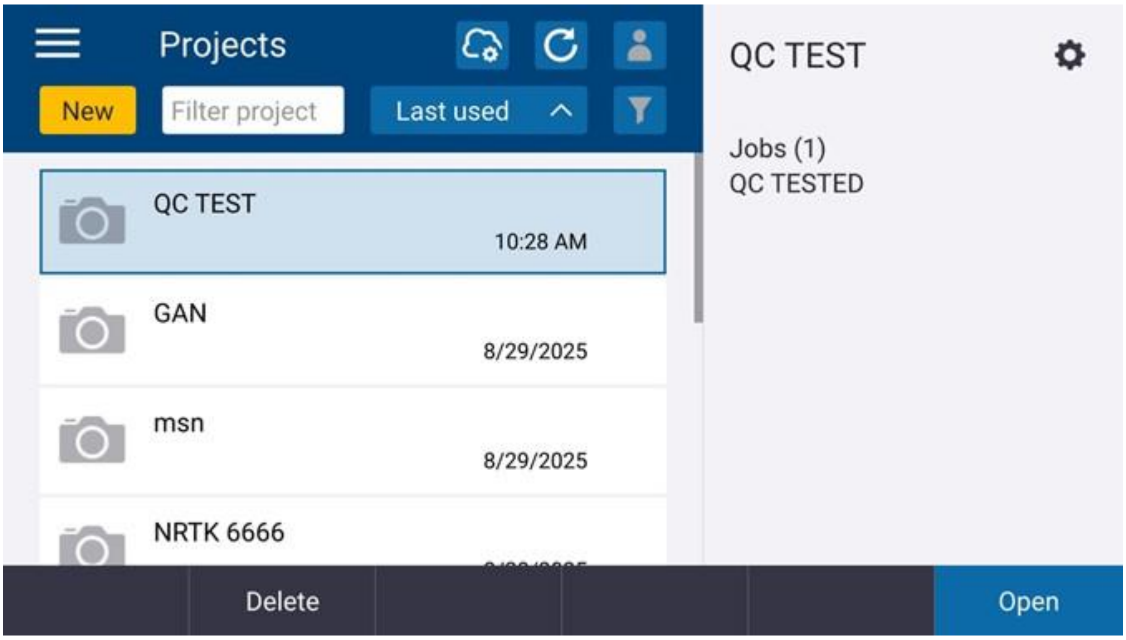
8.2 Setting up Rover

- Mount Rover Receiver R980 on Survey Pole using Quick Release attachment.
- Measure the height of Survey Pole.
- Center the Rover at desired observation point & ensure levelling bubble is in center (In tilt situation it is not required).
- Switch on Rover Receiver.
- Do e-bubble calibration & IMU alignment **(see Annexure-A)**

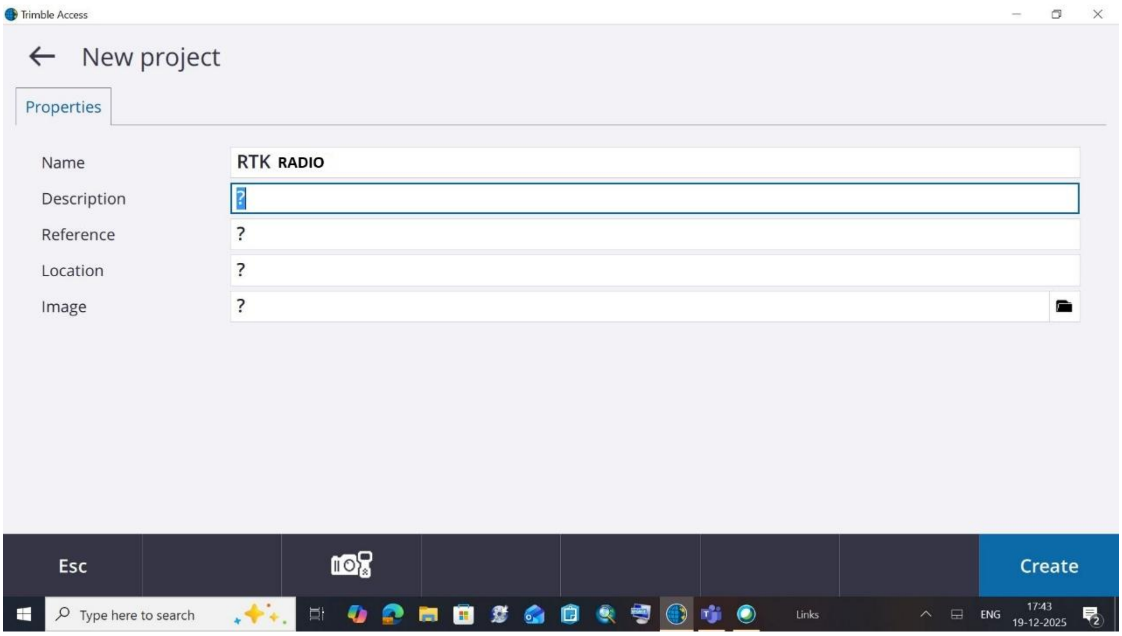
9. Procedure in Trimble Access

9.1 Create New Project & Job

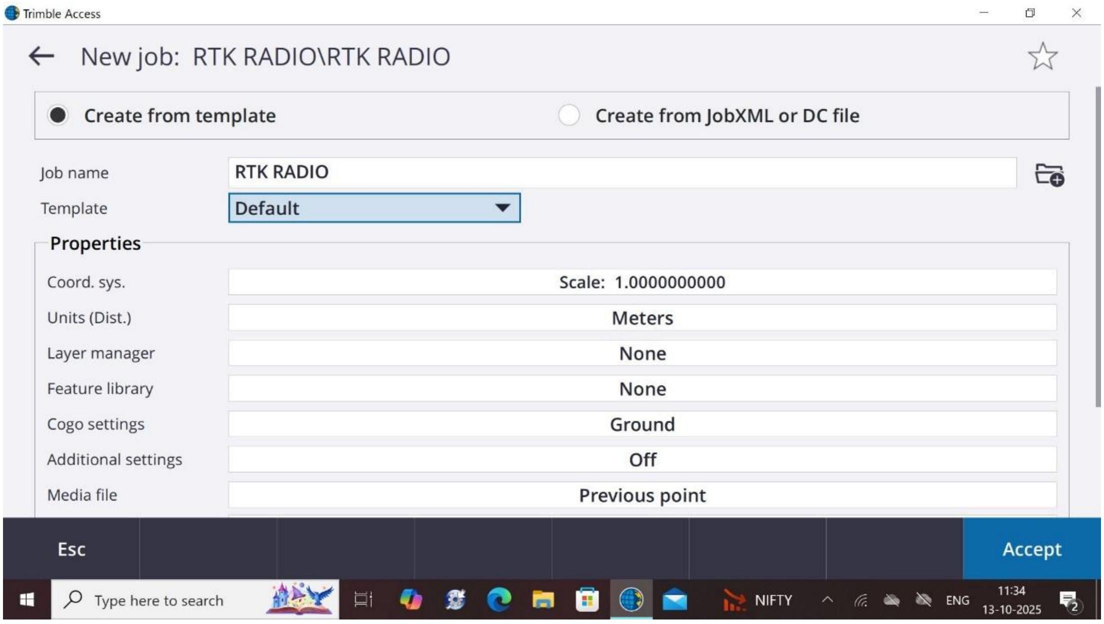
9.1.1 Open Trimble Access in Controller TSC5 or Getac, click on **New** to create a New Project



9.1.2 Give Project Name e.g. "RTK RADIO" and click >>**Enter**>> **Create**



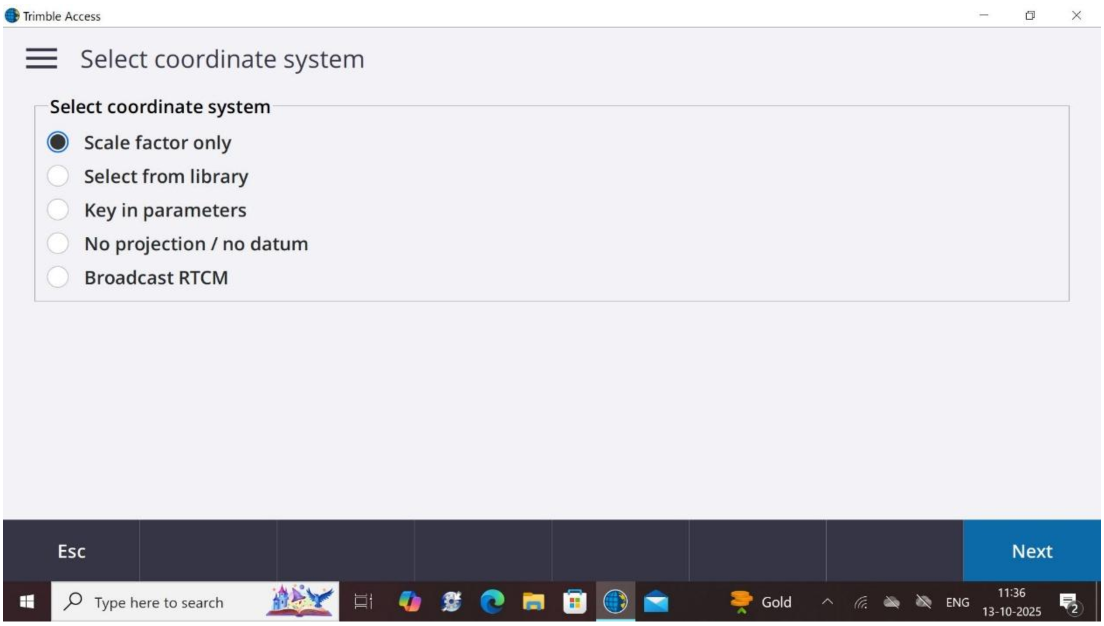
➢ Inside project create job, Give a name to this new job e.g. "RTK RADIO"



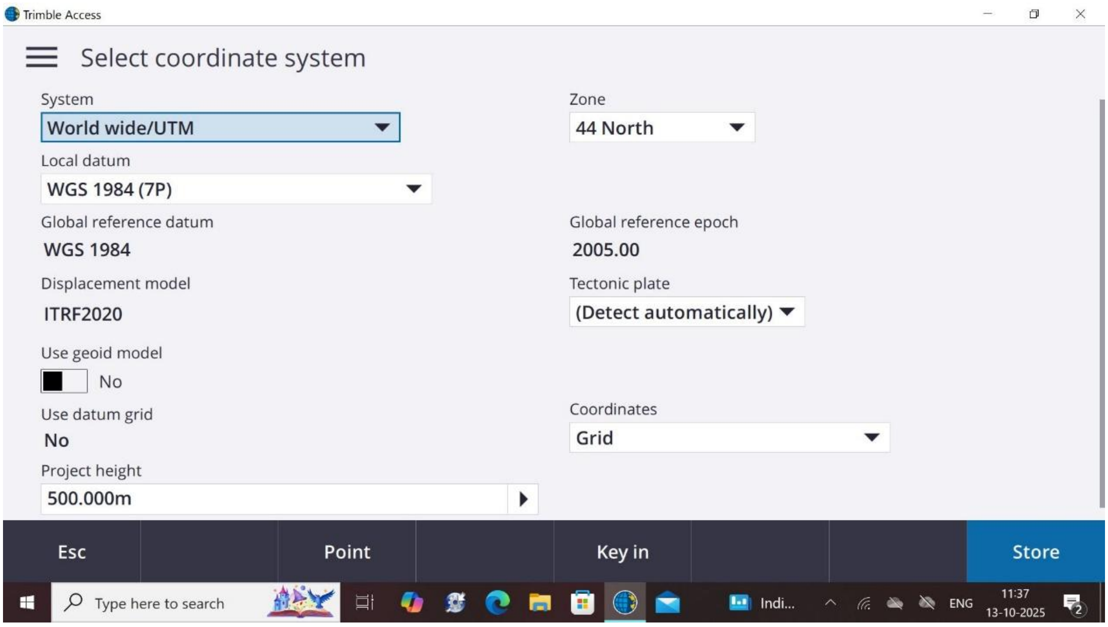
9.2 Set Coordinate System, Units & CSV File Name

➢ Click on **Coord.sys.**

- Then Click on **Select from library**

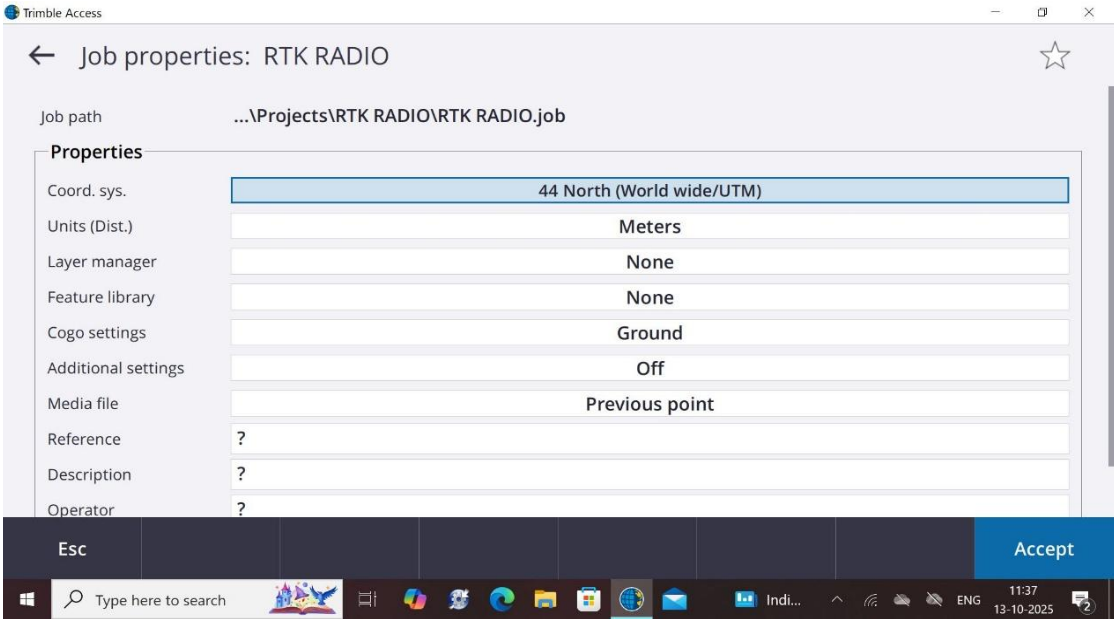


- In the **Select coordinate system**
- Select System: World wide/UTM from drop down menu
- Select Zone: Calculate and select from dropdown menu as per your working area Longitude (44N for Hyderabad)
- Select Local datum: WGS 1984 (7P) from drop down options
- Global reference datum: WGS 1984 (Default)
- Global reference epoch: 2005(Default)
- Displacement model: ITRF2020 (Default)
- Tectonic plate: Detect automatically
- Use Geoid model Click YES (if any local Geoid model is available) & select appropriate GGF file or else select NO.

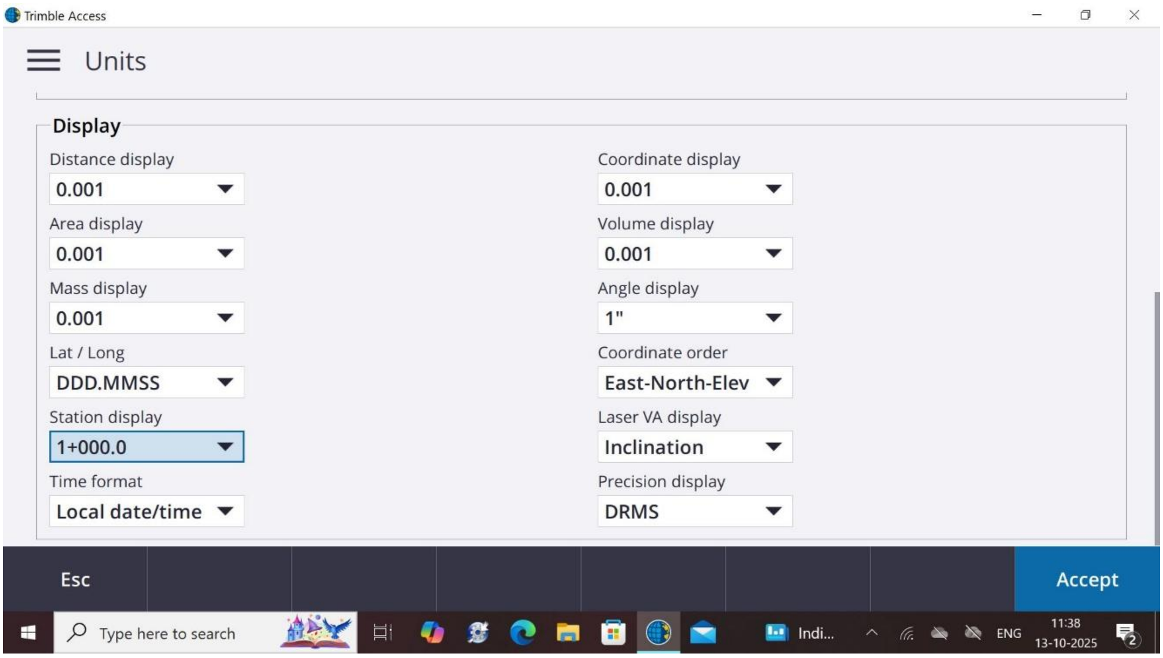


- Use datum grid: No
- Coordinates: Grid.
- Project Height: Give average Height of the working area.
- Click >>**Store**

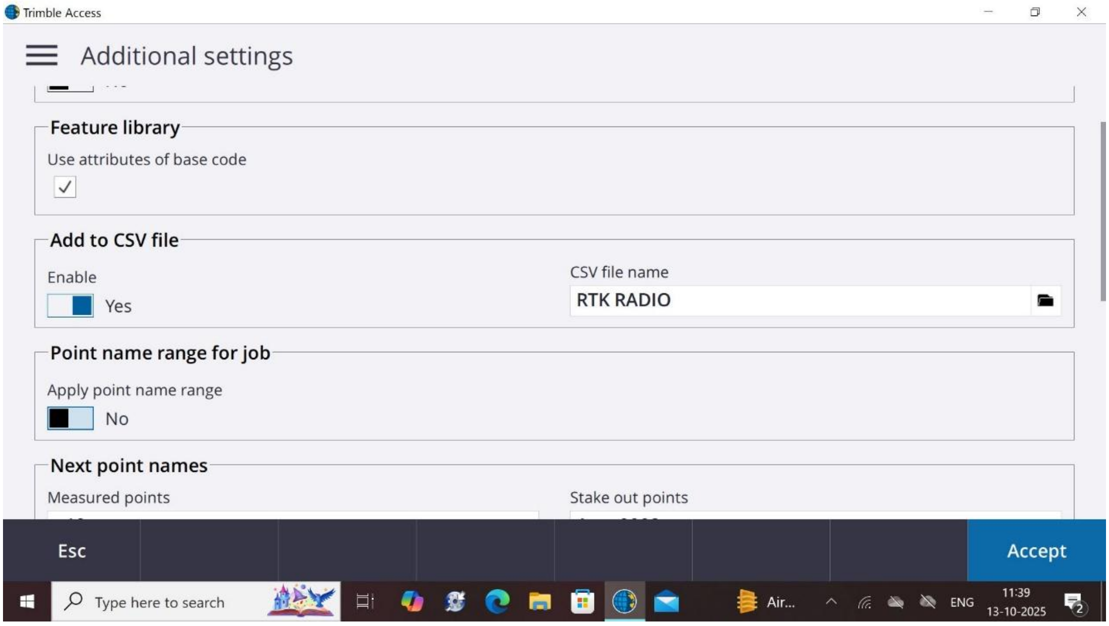
➢ Now click on **Units(Dist)**



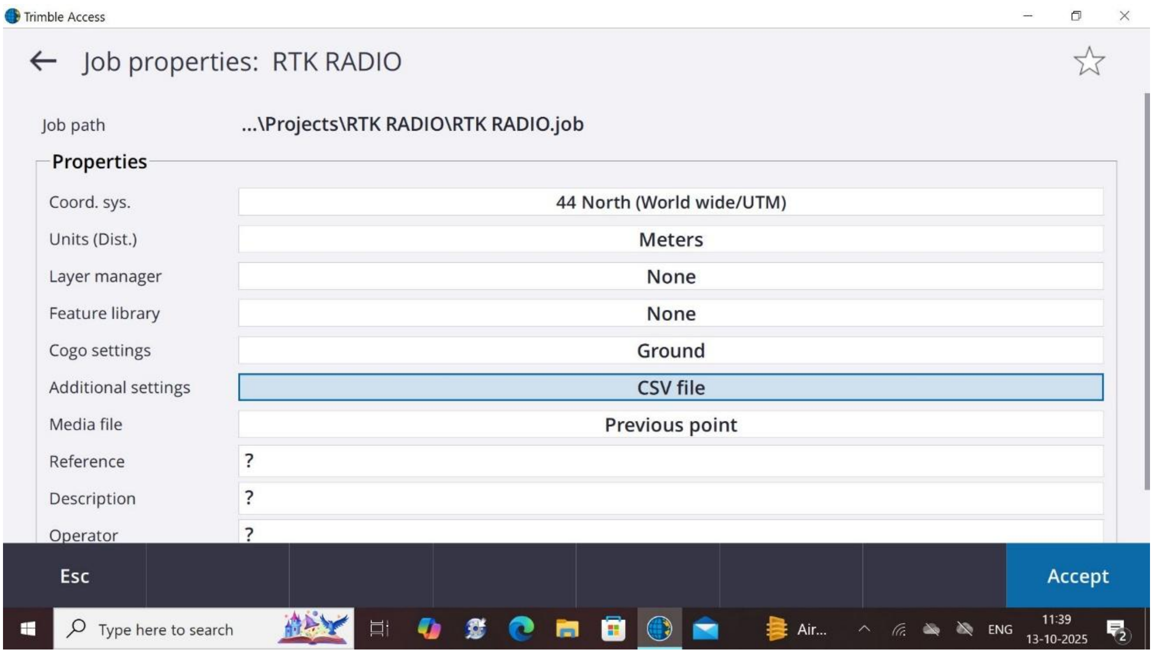
➢ Select options from following window as per convenience and Click on **Accept**



➢ Now again Click>> **Additional settings**



➢ **Enable** Add to csv & give name e.g."RTK RADIO" to .csv file



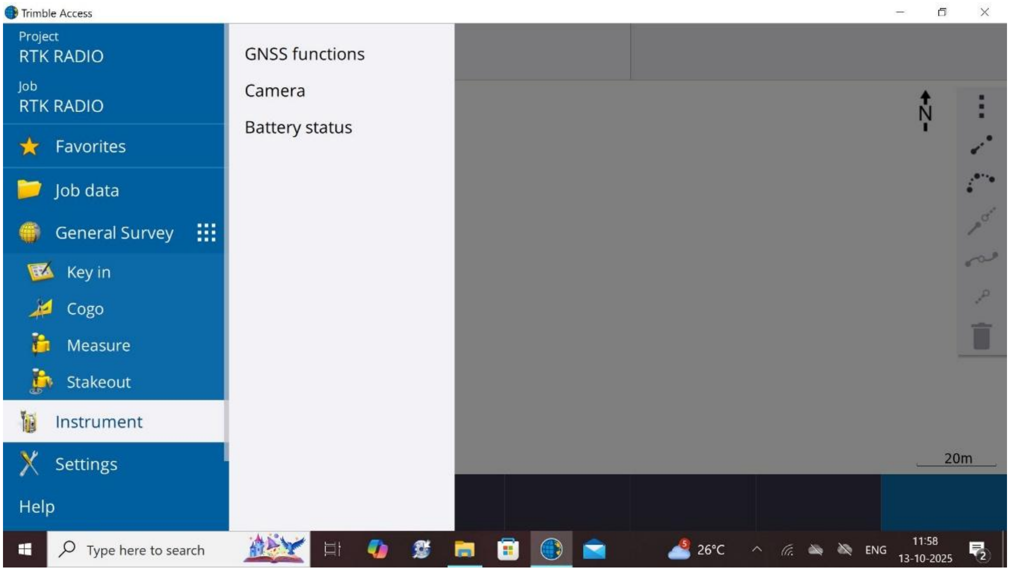
➢ Click >> **Accept**

Now a New Project & a New Job with a defined projection system has been created.

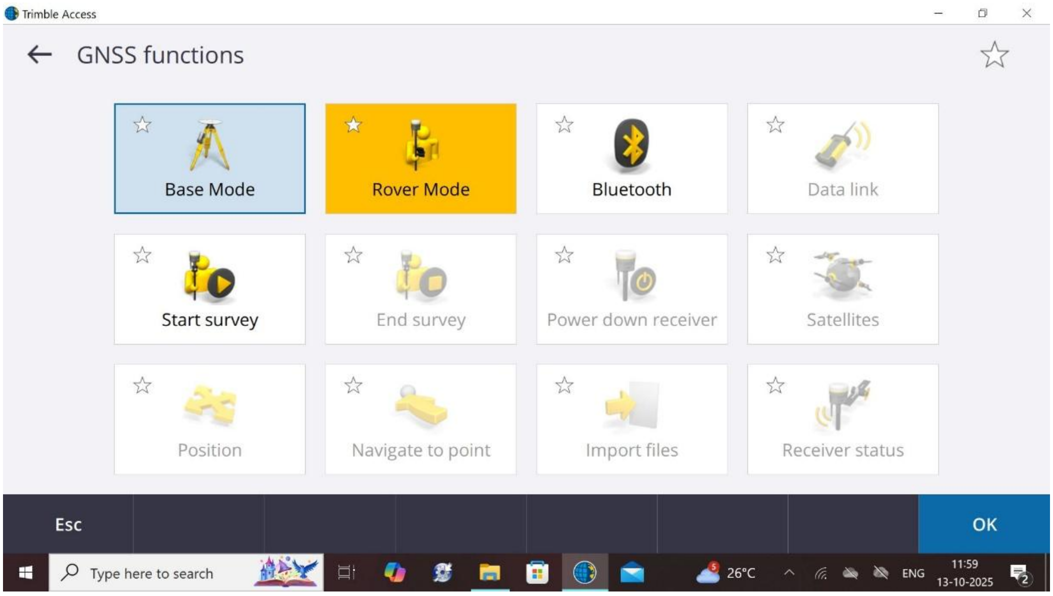
10. Configure Base & Rover Receiver with Controller

In order to configure Base & Rover Receiver with Controller

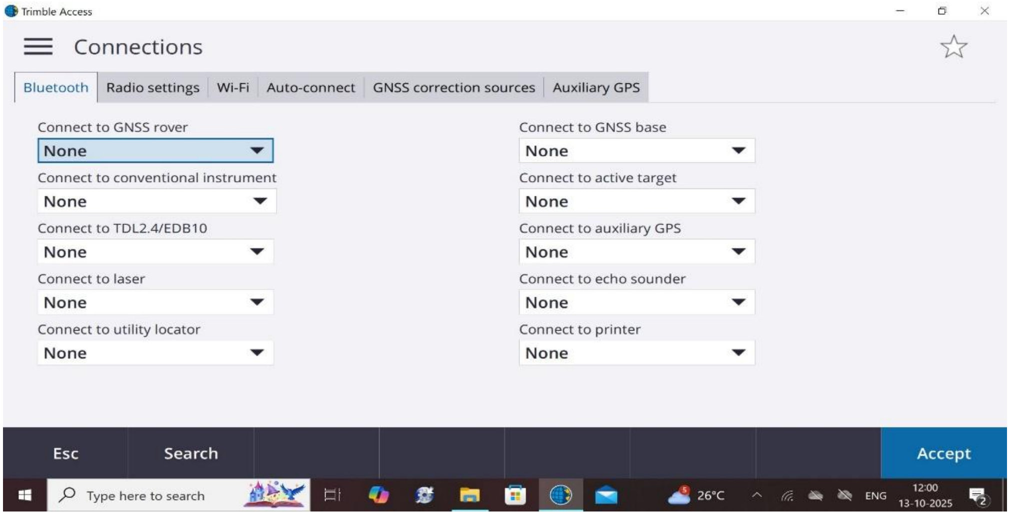
➢ Click on **Instruments** >> **GNSS Functions**



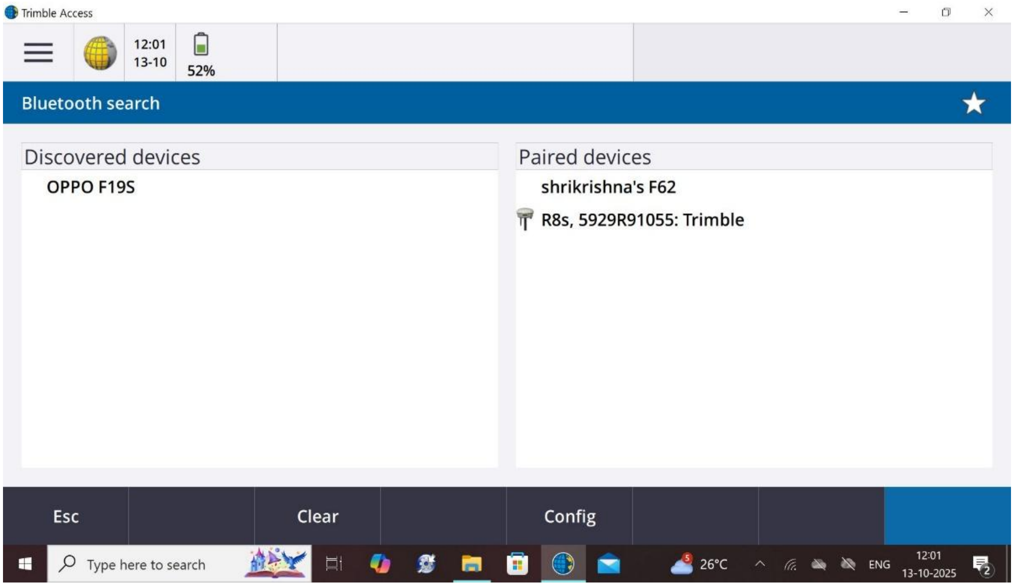
➢ Click on **GNSS Functions**>> **Bluetooth**

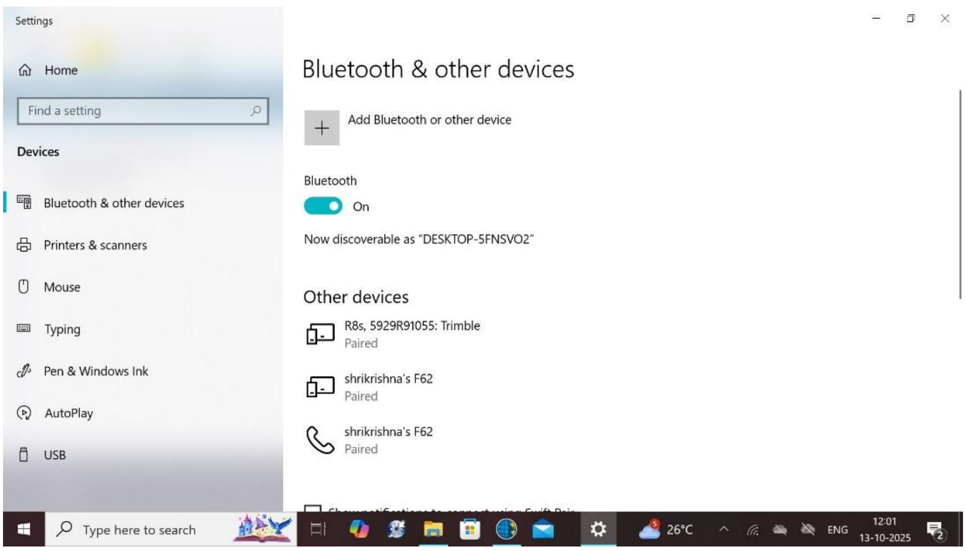


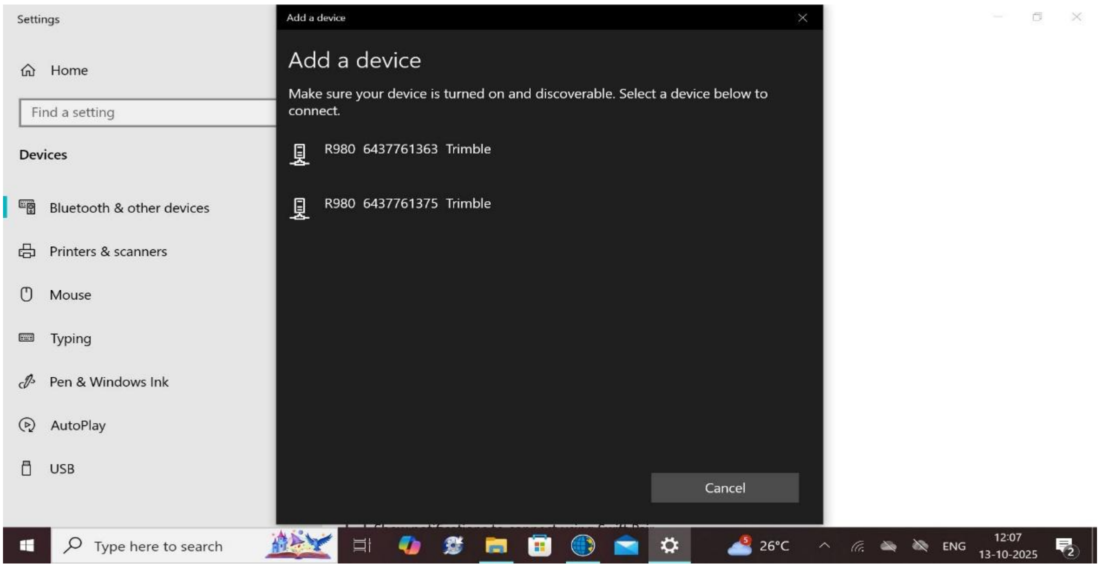
- Now Click on **Search**

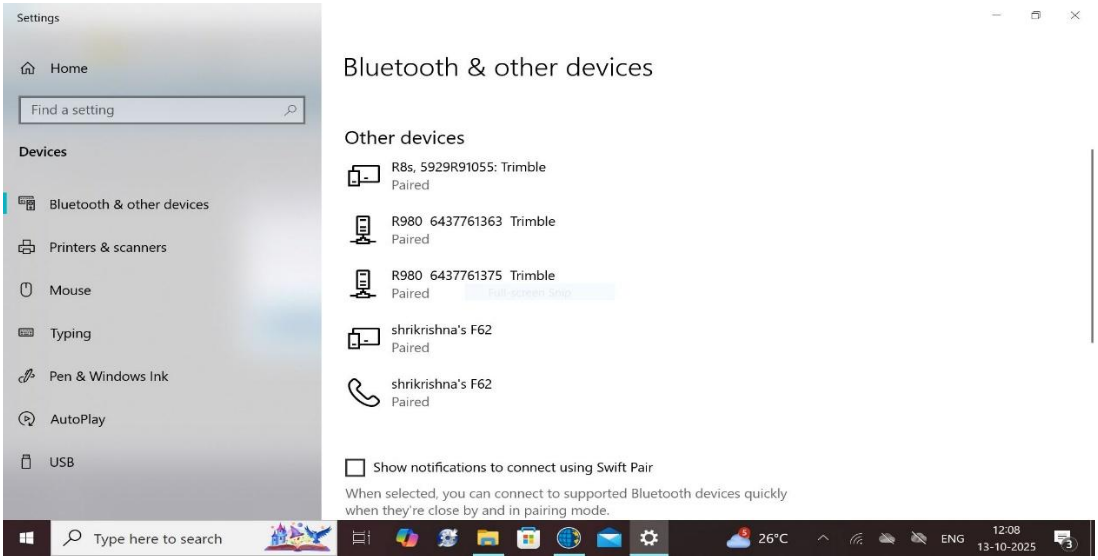


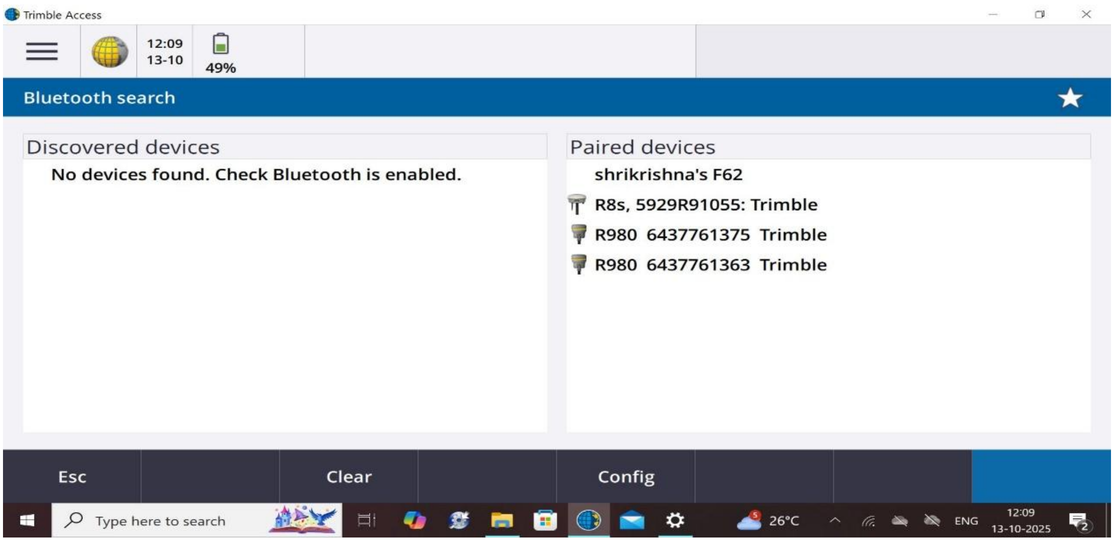
- Click on **Config** and Pair Base & Rover Receiver as per following steps:



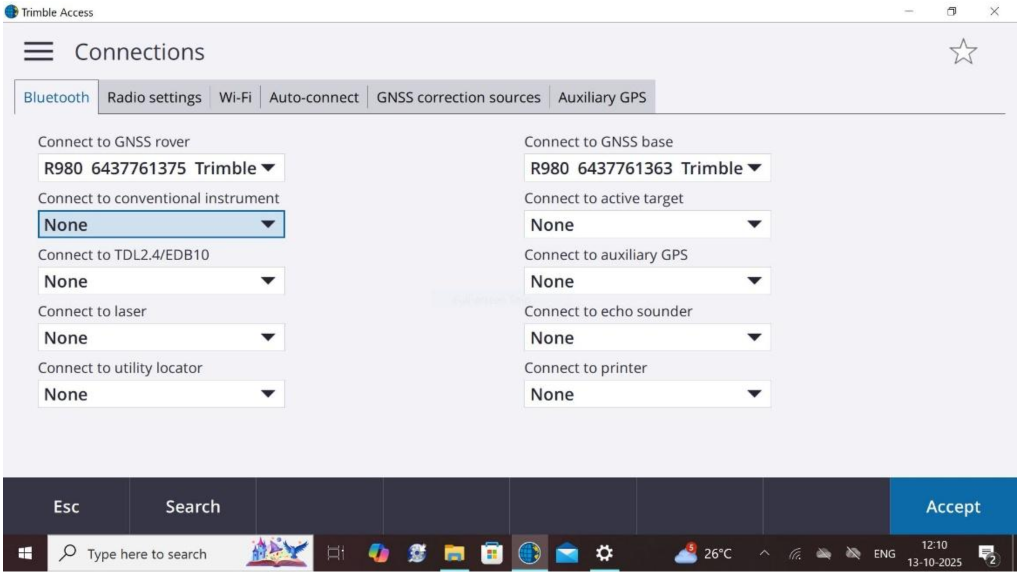



Settings
Add a device
Add a device
Make sure your device is turned on and discoverable. Select a device below to connect.
R980 6437761363 Trimble
R980 6437761375 Trimble
Cancel
Home
Find a setting
Devices
Bluetooth & other devices
Printers & scanners
Mouse
Typing
Pen & Windows Ink
AutoPlay
USB
Type here to search
26°C
ENG
12:07
13-10-2025


Settings
Home
Find a setting
Devices
Bluetooth & other devices
Printers & scanners
Mouse
Typing
Pen & Windows Ink
AutoPlay
USB
Bluetooth & other devices
Other devices
R8s, 5929R91055: Trimble
Paired
R980 6437761363 Trimble
Paired
R980 6437761375 Trimble
Paired
shrikrishna's F62
Paired
shrikrishna's F62
Paired
Show notifications to connect using Swift Pair
When selected, you can connect to supported Bluetooth devices quickly when they're close by and in pairing mode.
Type here to search
26°C
ENG
12:08
13-10-2025


Trimble Access
12:09
13-10
49%
Bluetooth search
Discovered devices
No devices found. Check Bluetooth is enabled.
Paired devices
shrikrishna's F62
R8s, 5929R91055: Trimble
R980 6437761375 Trimble
R980 6437761363 Trimble
Esc
Clear
Config
Type here to search
26°C
ENG
12:09
13-10-2025

➢ Select respective Receiver from Paired Receivers as follows:



Click >> **Accept**

11. Creating Survey Styles

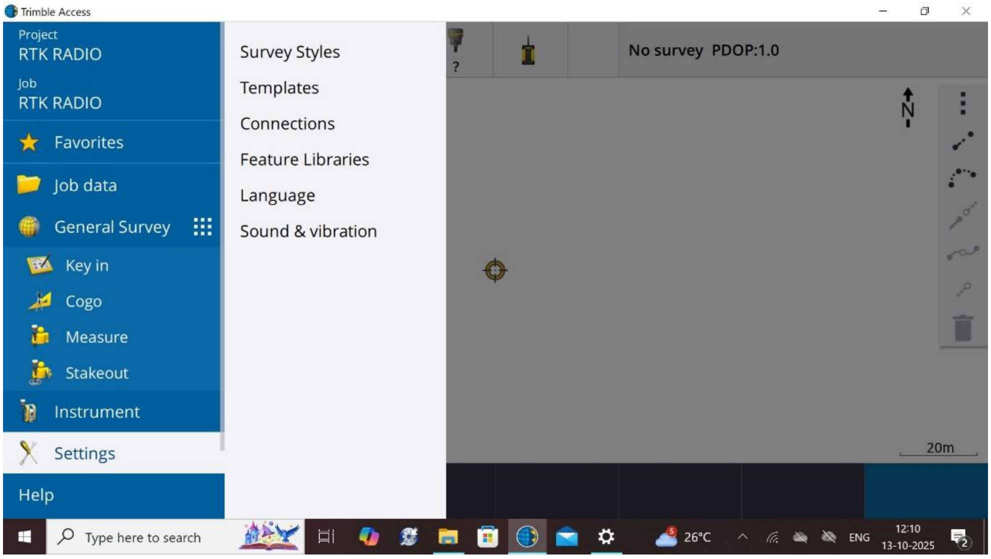
In order to create a Survey Style in RTK Method we have to define following things:-

- Rover Option
- Rover Data Link
- Base Option
- Base Data Link

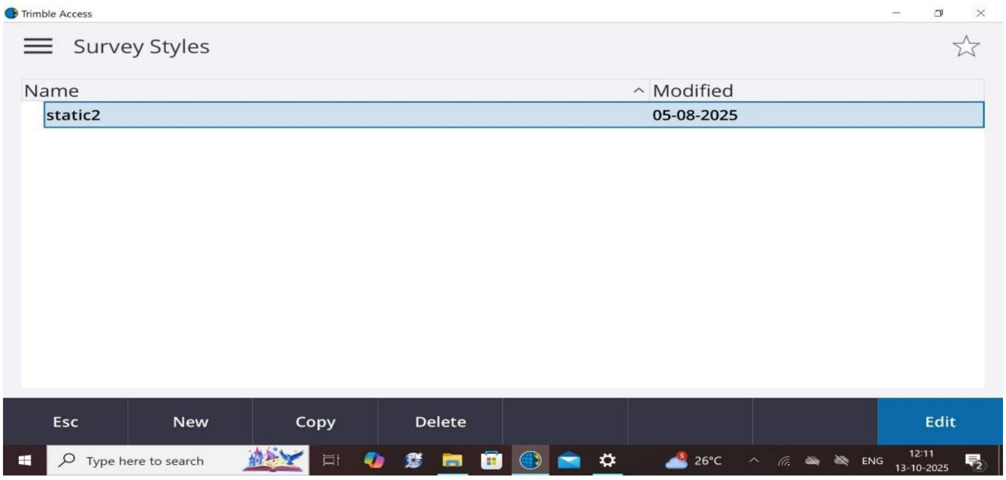
Following are the steps mentioned to complete above settings.

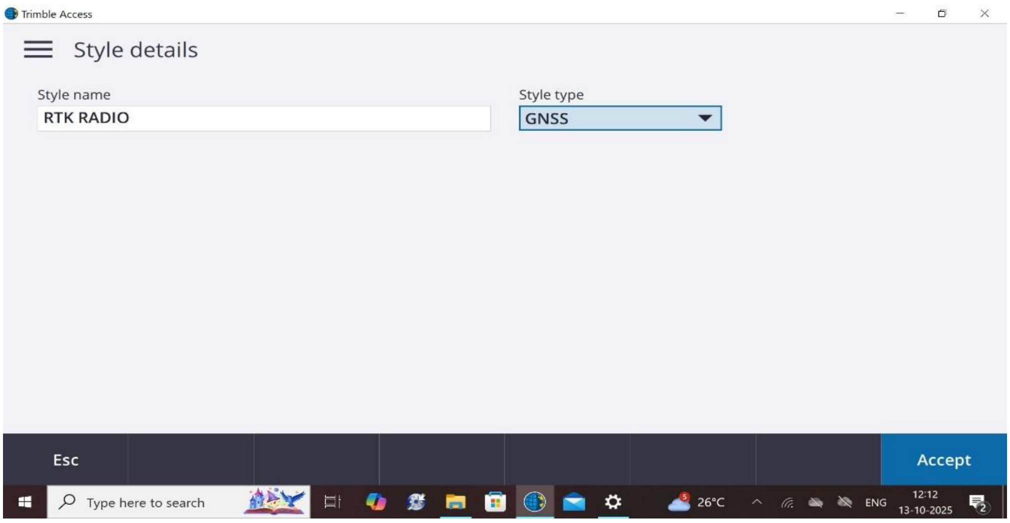
➢ Create a New Survey style for RTK method, for that click main menu

➢ Go to **Settings** >> **Survey Styles**



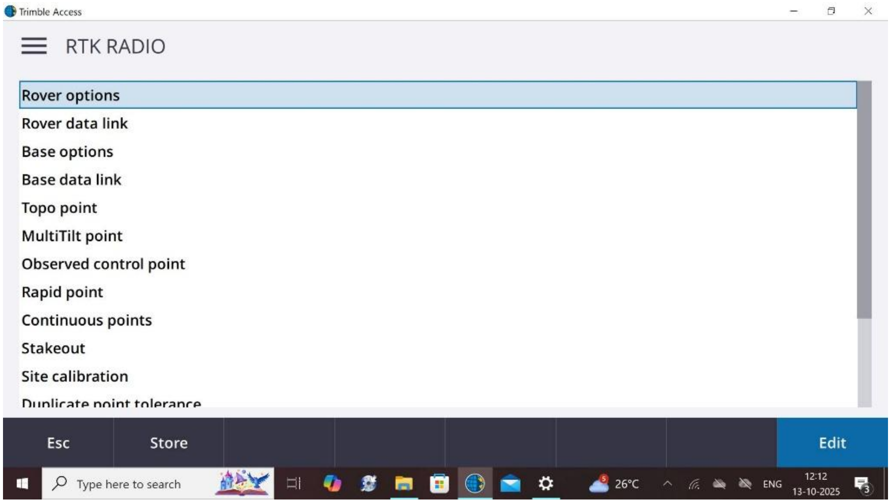
➢ Click on **New** and Give name of style e.g. RTK RADIO and **Accept.**





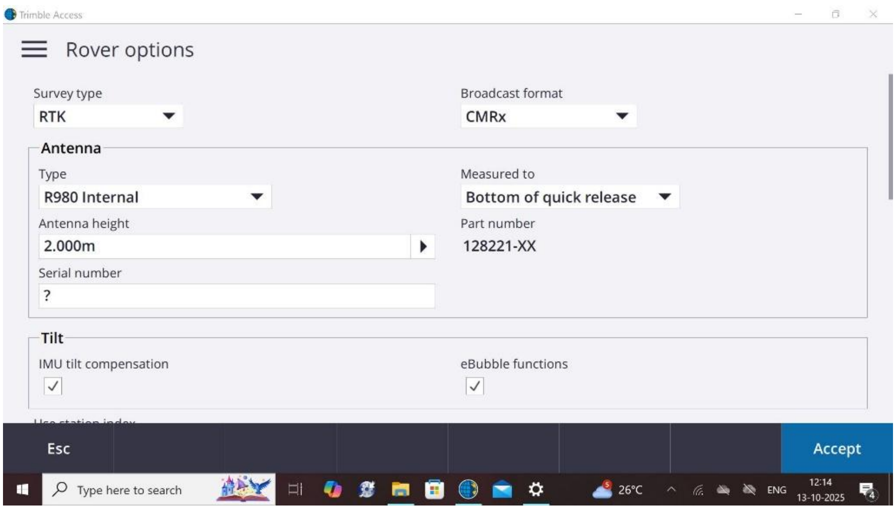
A new window will open as follows:

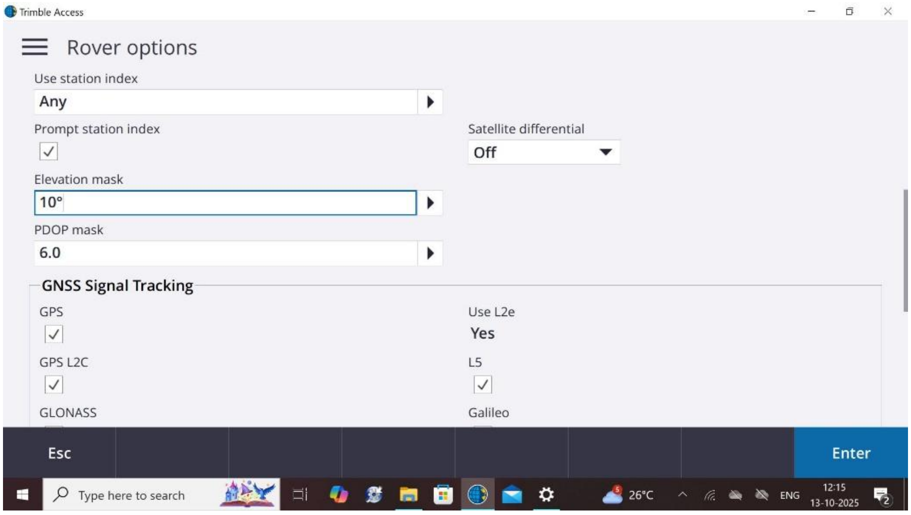
➢ IN Rover Options click on **Edit**

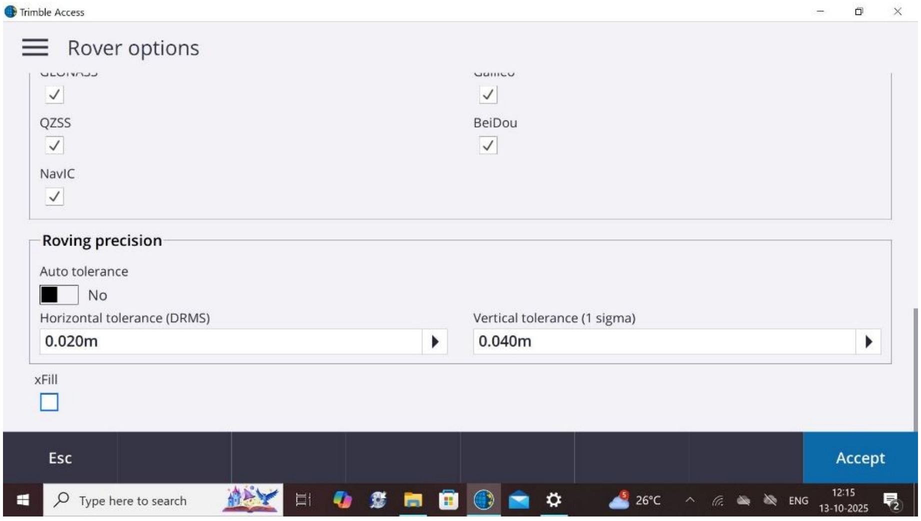


In Rover Options

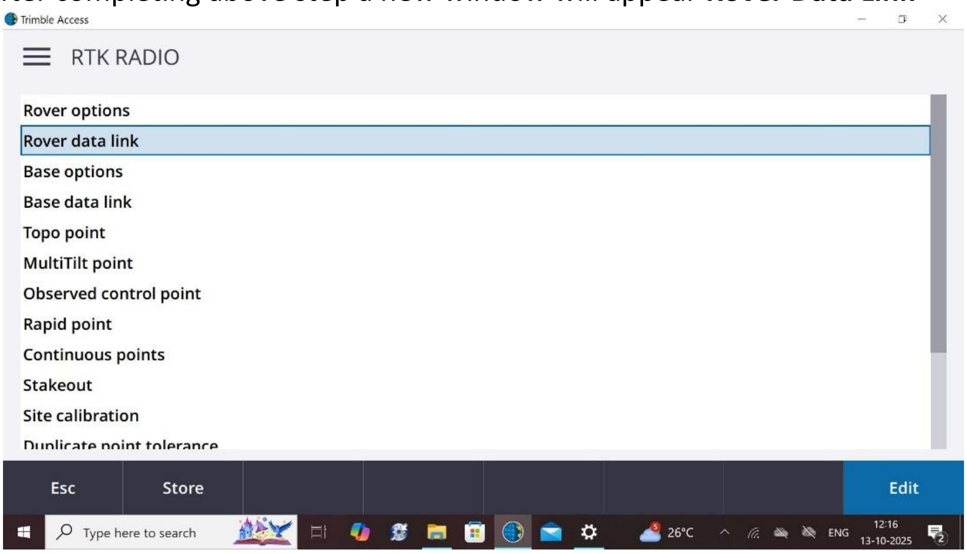
- Select Survey Type as RTK
- Broadcast Format: CMRx
- Provide specifications of GNSS Antenna using.
- Check on IMU tilt compensation & eBubble Functions if desirable.
- Keep Satellite Differential Off.
- Enter Elevation Mask angle, PDOP value.
- Check on all GNSS Constellation.
- Enter Roving precision values in Horizontal & Vertical as per requirements.
- Check on xFill if it is desirable.







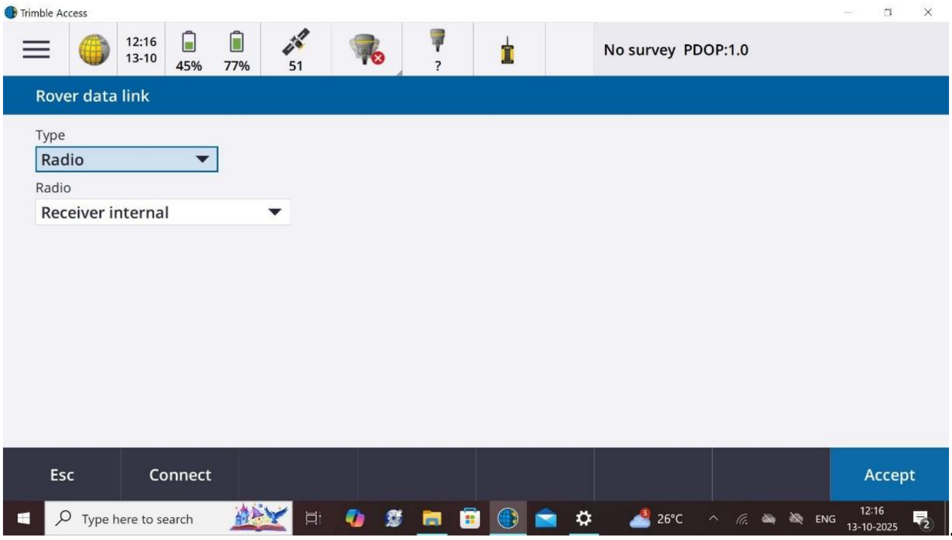
- ➢ Click on **Accept.**
- ➢ After completing above step a new window will appear **Rover Data Link**



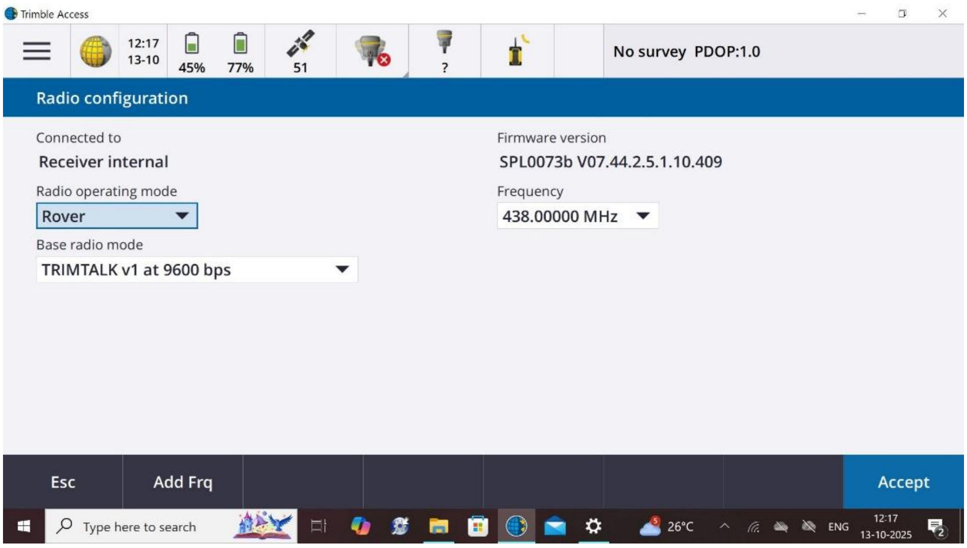
- ➢ Click on **Edit**

➢ In **Rover data link** Tab we have to define the way by which Rover would be connected to Base. Here we are using Internal Radio of Rover Receiver.
Select Radio type from dropdown list i.e. **Receiver Internal** and click on **Connect.** In subsequent window we have to select Radio operating mode as **Rover** ; Frequency **438 MHz o/ 441 MHz** & **Base radio mode >> 9600bps** (bits per second).

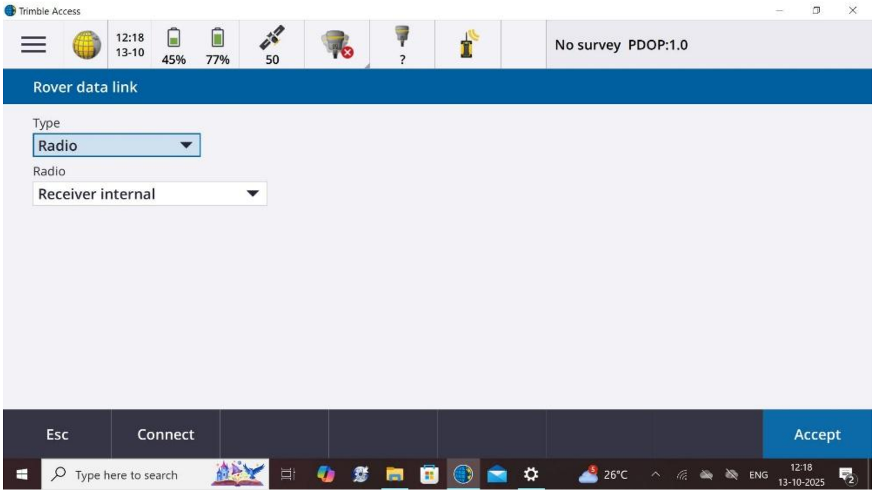
Note:- 9600 bps is the standarad rate of data transmission it can be changed as per requirements or available options.
If Frequency is selected 438 MHz or 441 MHz , same frequency should be selected while defining Base Data Link.



➢ Click on **Accept**



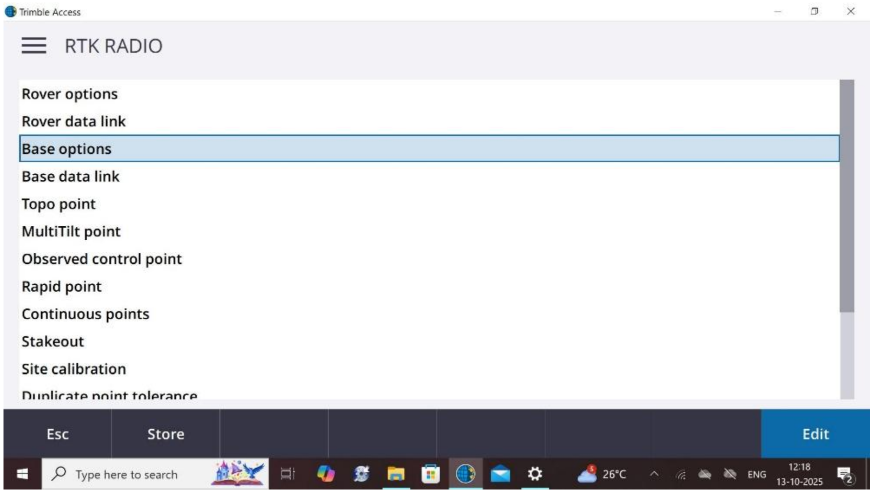
➢ Again Click on **Accept**

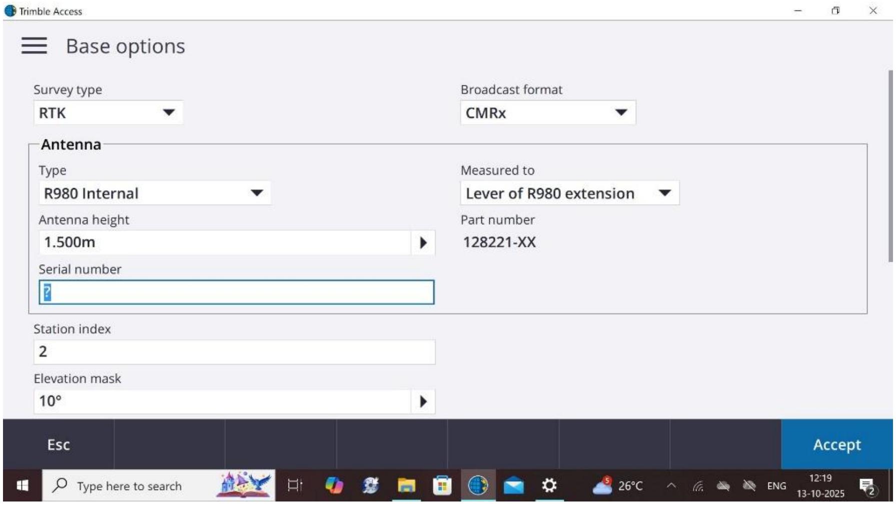


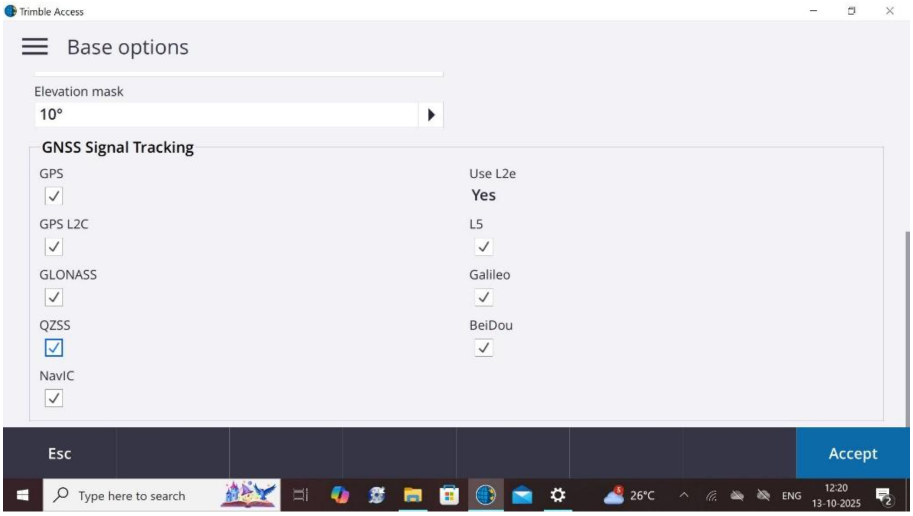
➢ Now click >> **Base options** >> **Edit**

In Base options

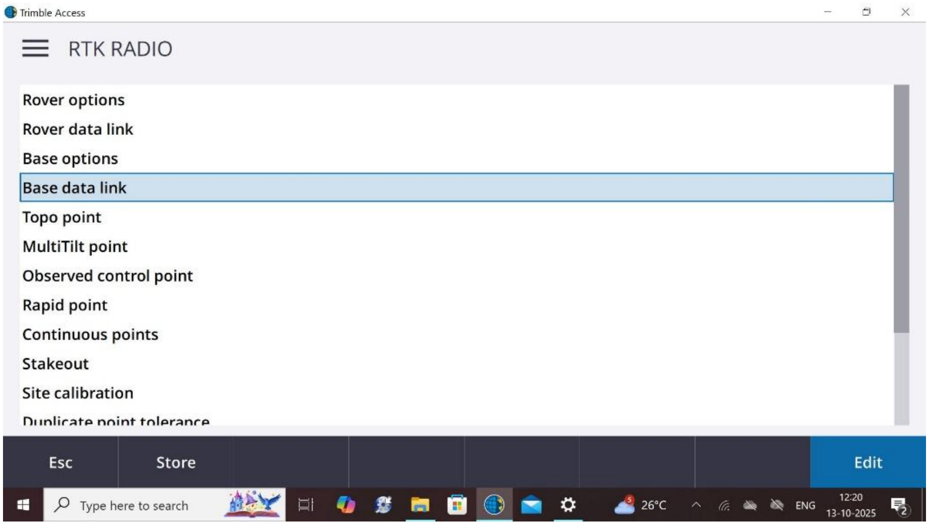
- Select Survey Type as RTK
- Broadcast Format: CMRx
- Provide specifications of GNSS Antenna using.
- Insert Station Index No.(any from 1 to 31)
- Enter Elevation Mask angle, PDOP value.
- Check on all GNSS constellations.







- ➢ Click >> **Accept**
- ➢ After completing above step a new window will appear **Base Data Link**



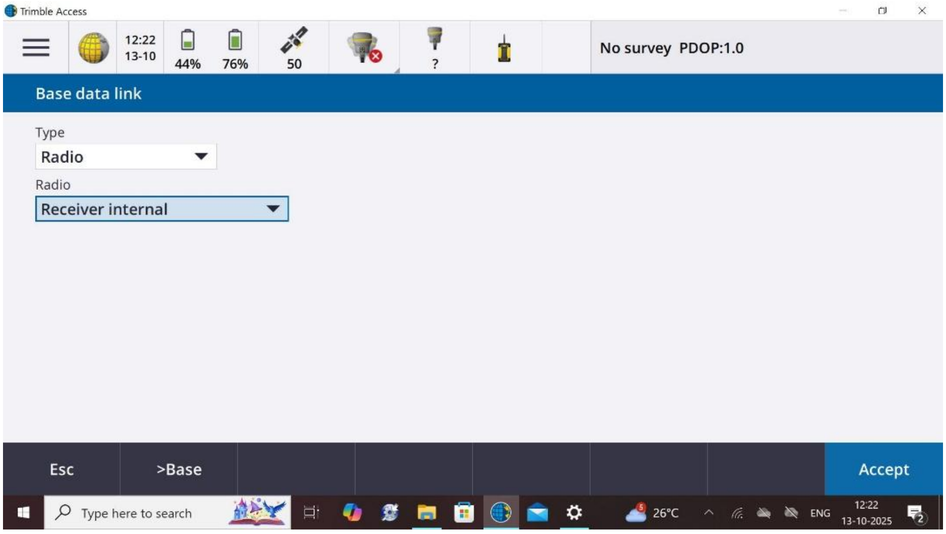
- ➢ Click on **Edit**

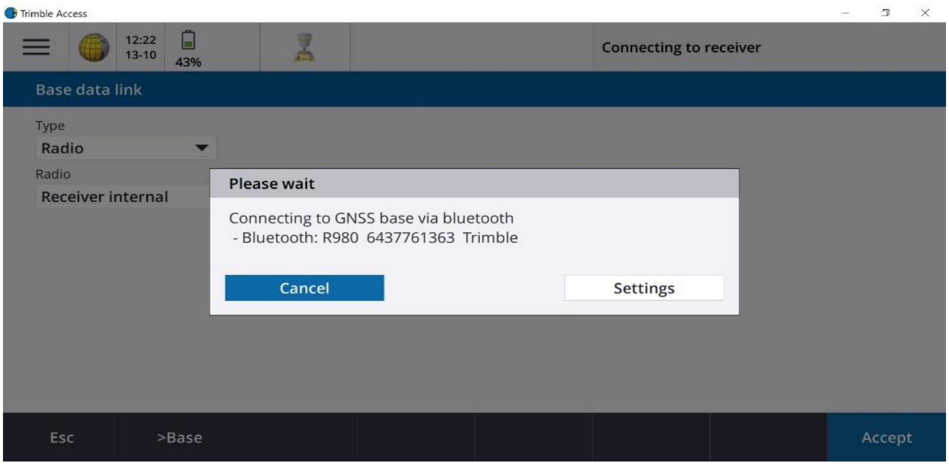
➢ In **Base data link** Tab we have to define the way by which Base would be connected to Receiver. Here Base can be connected to Receiver via Internal Radio Or External Radio.

11.1 Case 1. Using Internal Radio of Base Receiver.

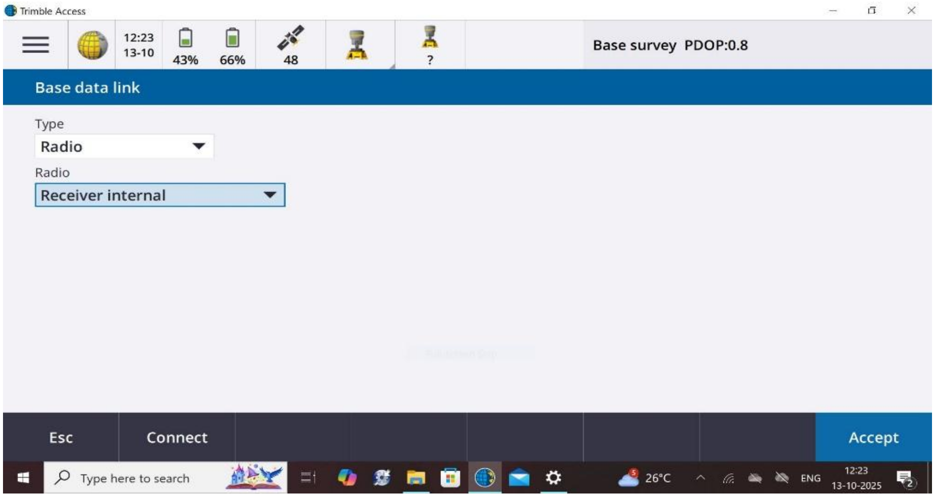
For this select **Type** >> **Radio** and **Radio** >> **Receiver Internal** from dropdown list and Click >> **Accept.**

➢ Now Click on **>Base**

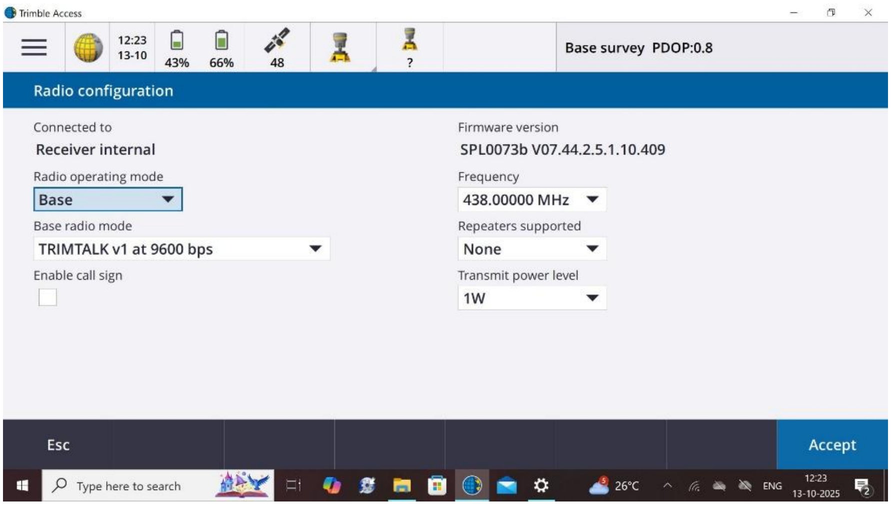




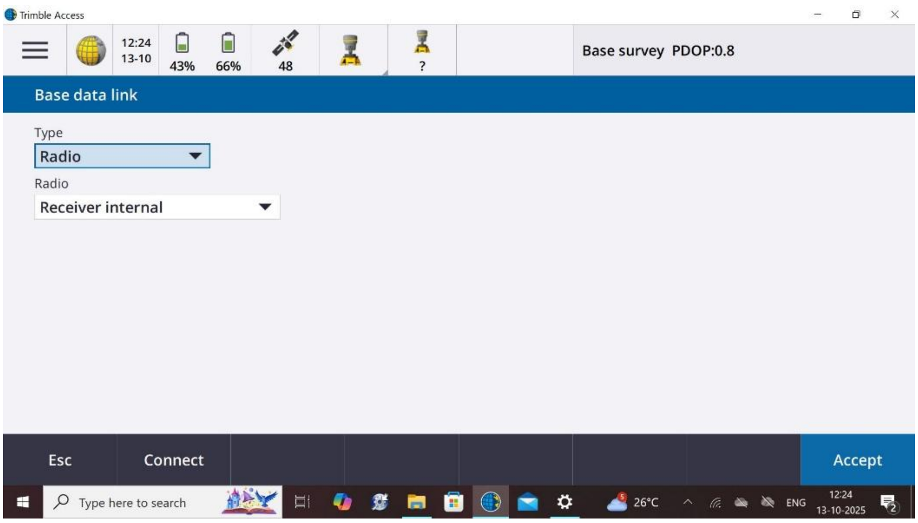
➢ Now Click >> **Connect**



- In upcoming window select **Radio operating mode** >> **Base ; Base Radio mode** >> **9600 bps ;Frequency** >> **438 MHz** (same as Rover Radio)**; Repeaters** >> **None** (if there is No Repeater) and **Transmit power level** >> **1W/0.5W**



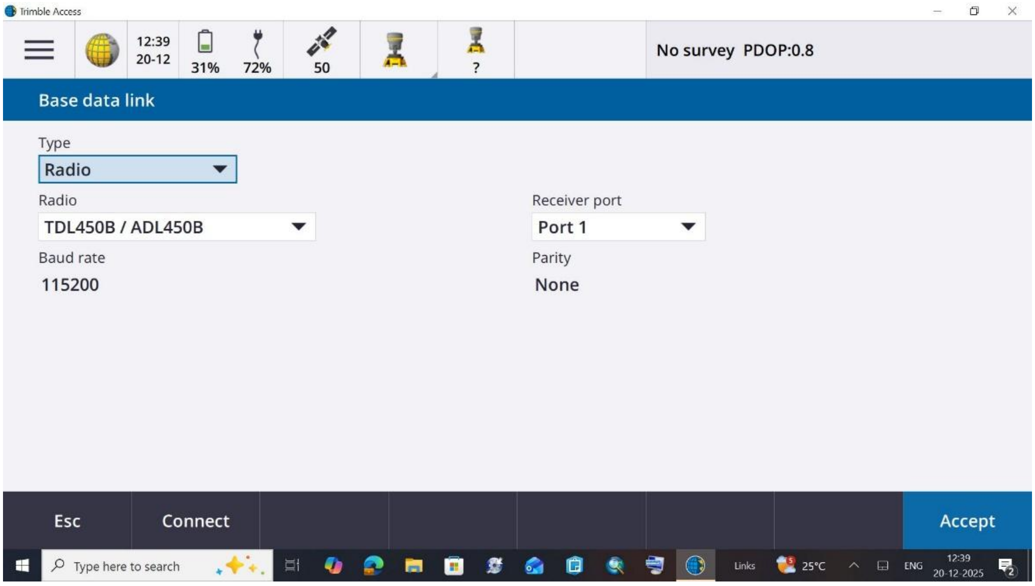
- Click >> **Accept**



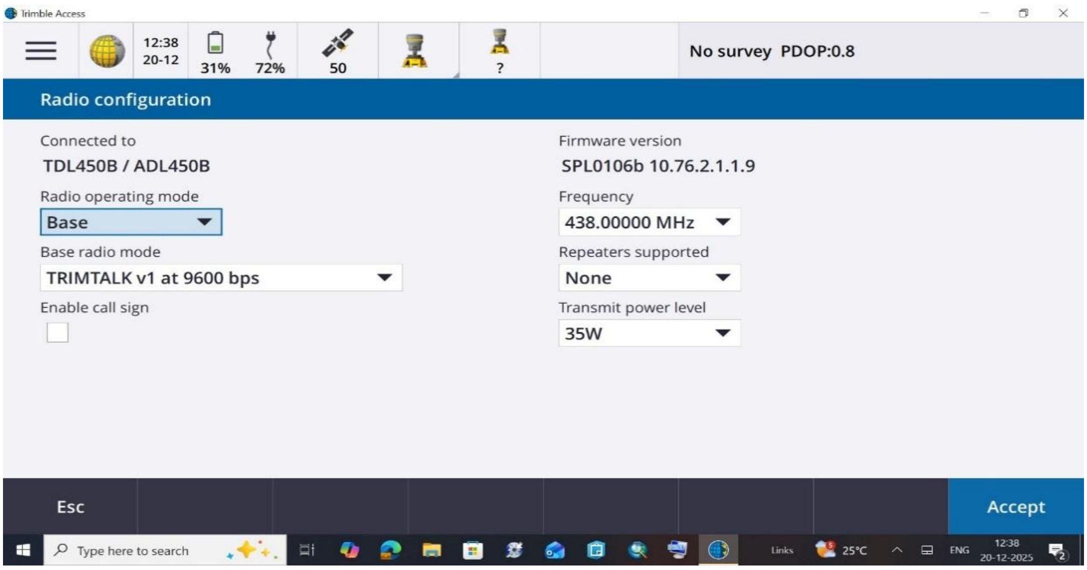
- Click >> **Accept**

11.2 Case 2. Using External Radio of Base Receiver.

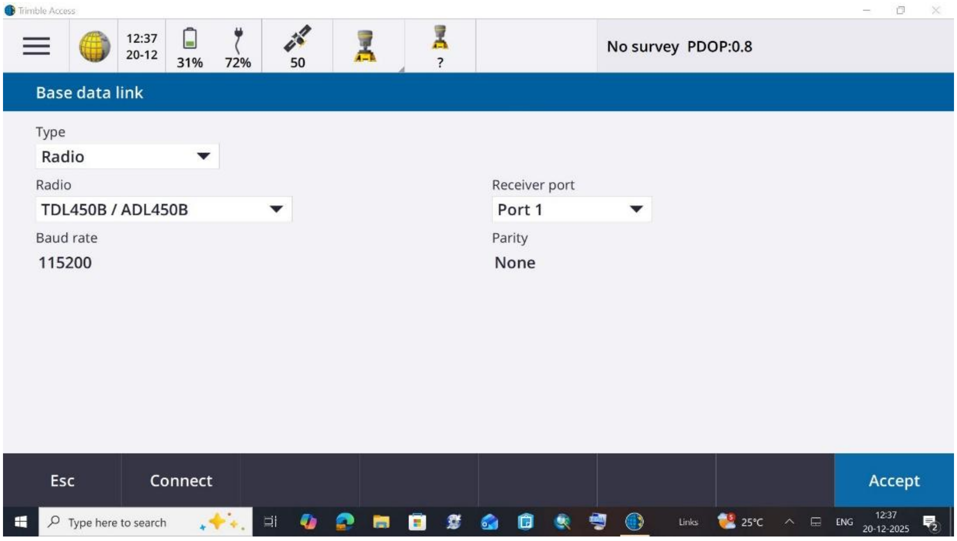
- For this select **Type** >> **Radio** ; **Radio** >> **TDL450B / ADL450B** ; **Receiver port** >> **Port1** from dropdown list and Click >> **Accept.**
- Now Click on **Connect**



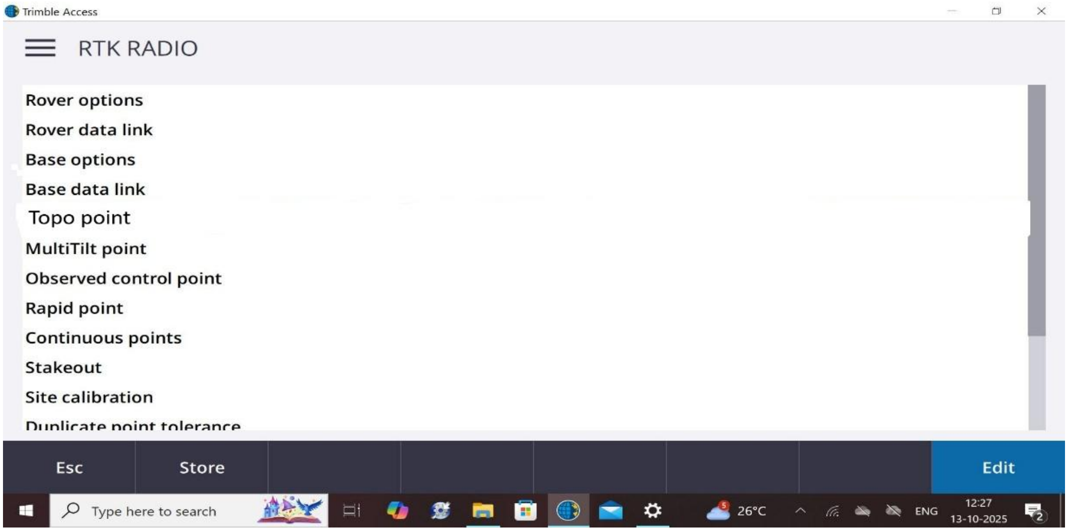
- In upcoming window select **Radio operating mode** >> **Base ; Base Radio mode** >> **9600 bps ;Frequency** >> **438 MHz** (same as Rover Radio)**; Repeaters** >> **None** (if there is No Repeater) and **Transmit power level** >> **35W**



- Click >> **Accept**



➢ Click >> **Accept**



➢ Click >> **Store**

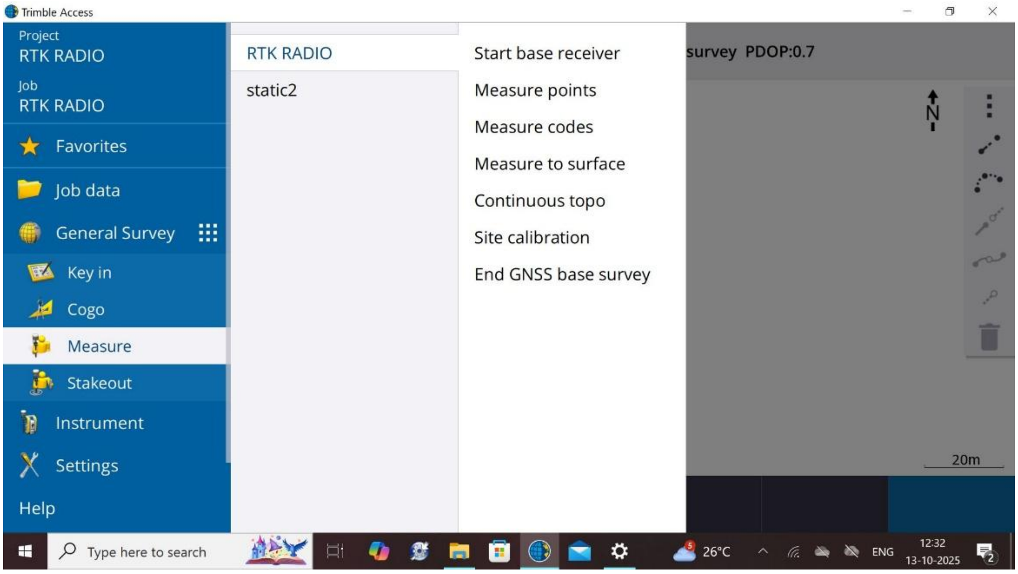
Using above steps our Base Receiver, Rover Receiver & Radios are configured and connected with Controller. Now in above window we can see other options also in which **Topo point**, **MultiTilt point, Observed Control point** & **Rapid point** are the methods of observing a point with different accuracies. **(see Annexure B)**

12. Field Procedure for RTK (with Radio)

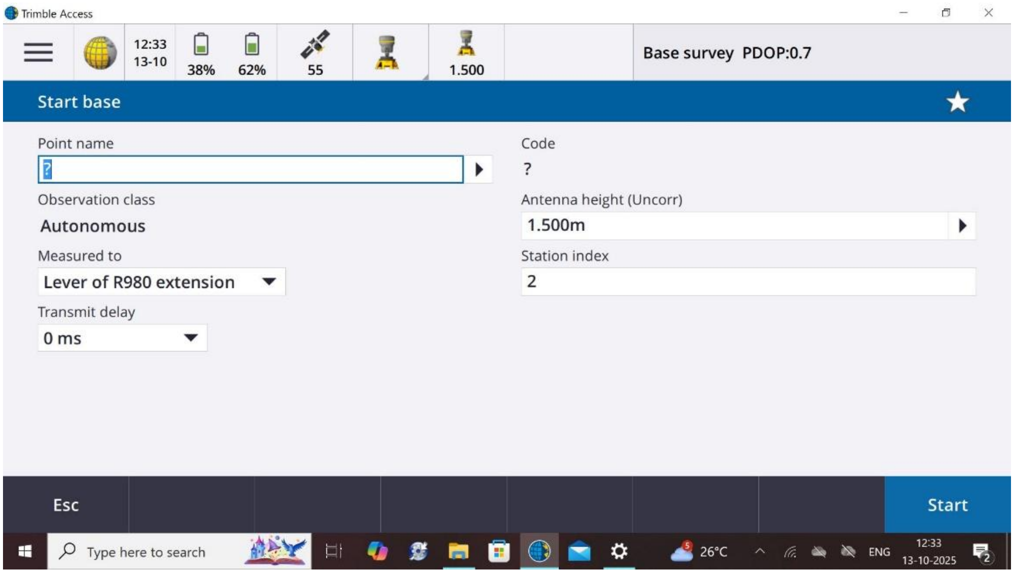
In the process of Field Observations using RTK method. It is completed in two steps, in First Step Base Receiver is started first and then Rover Receiver is stated.

12.1 Base Station initializing

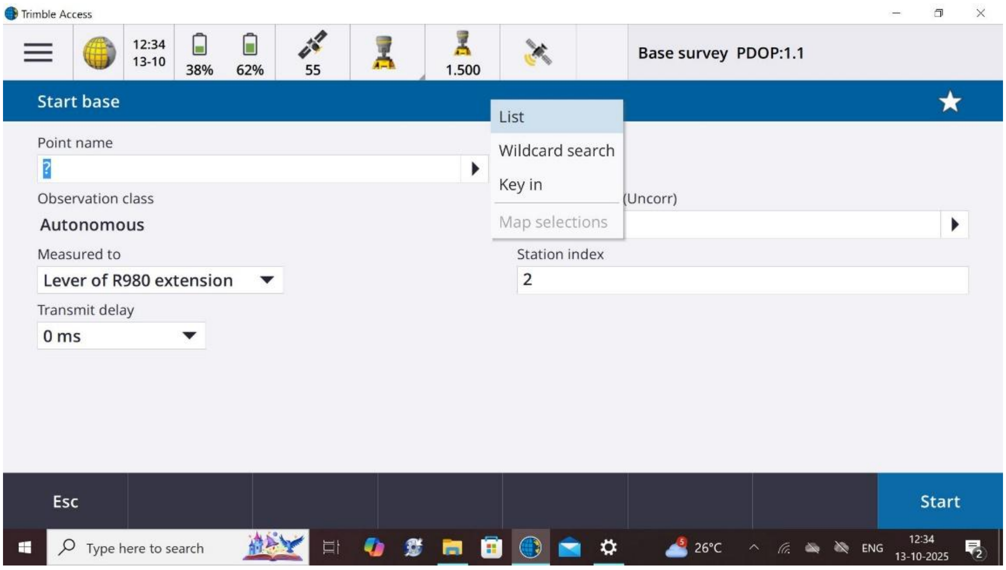
- Click on Menu >> **Measure** >> **RTK RADIO** (created survey style in earlier steps) >> **Start Base receiver**



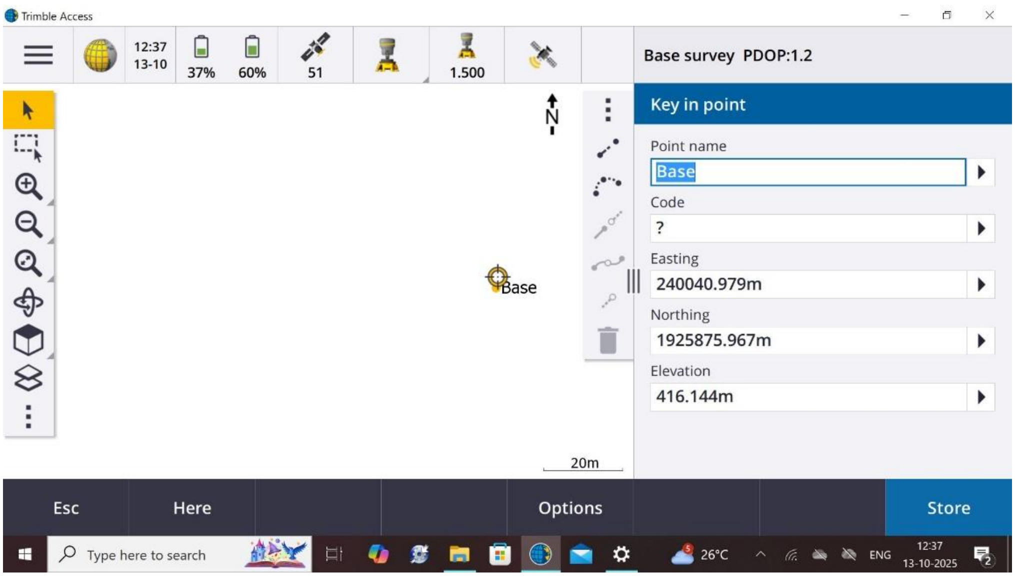
- In upcoming window again check the inserted values like Antenna Height, Station Index & Measure to options, also keep Transmit delay as 0 ms



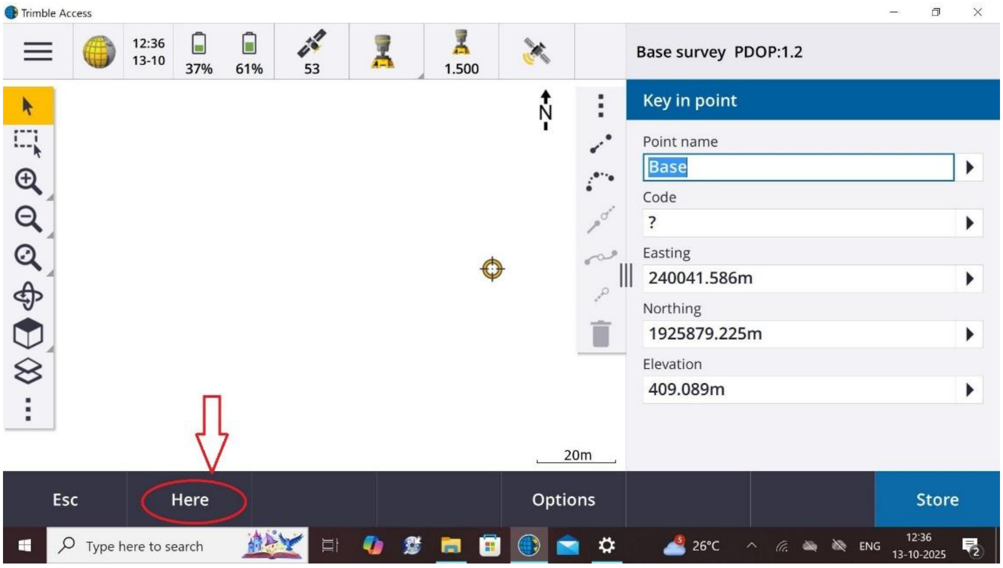
➢ Also in previous step Click on Option list after Point name, three options are shown in which List option is to choose a co-ordinates of Base station from already available List in Controller, Key-in option is to manually insert co-ordinates of Base station. Here we will Key-in manually.



➢ By Key-in option provide Point name and already known co-ordinates of Base station.

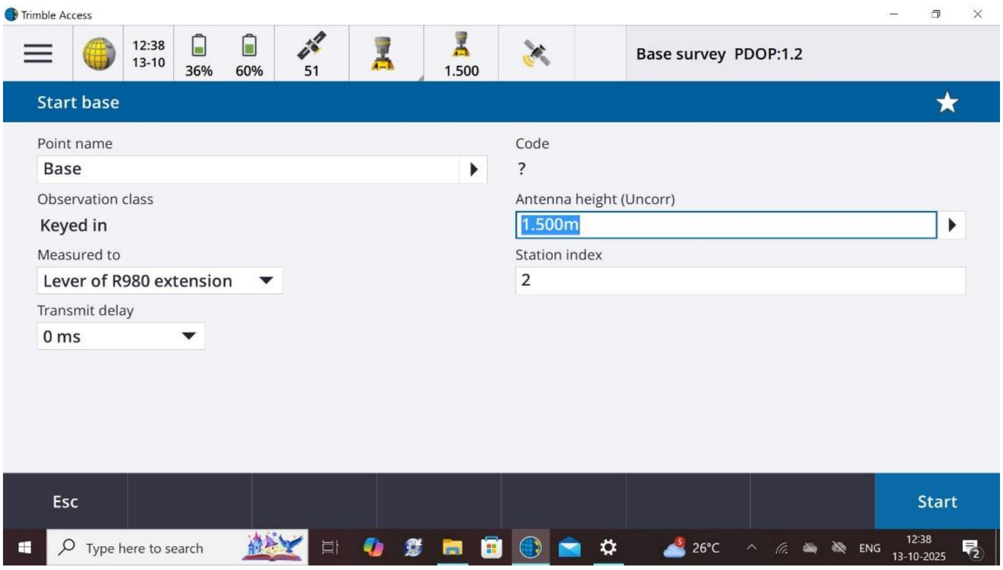


➢ If observer has not already known co-ordinates of Base station then there is a option **Here** on bottom, Click on it and Base Receiver will provide it's own un processed data of real time of Base. Which will be not same as known processed co-ordinates of Base in the previous step.

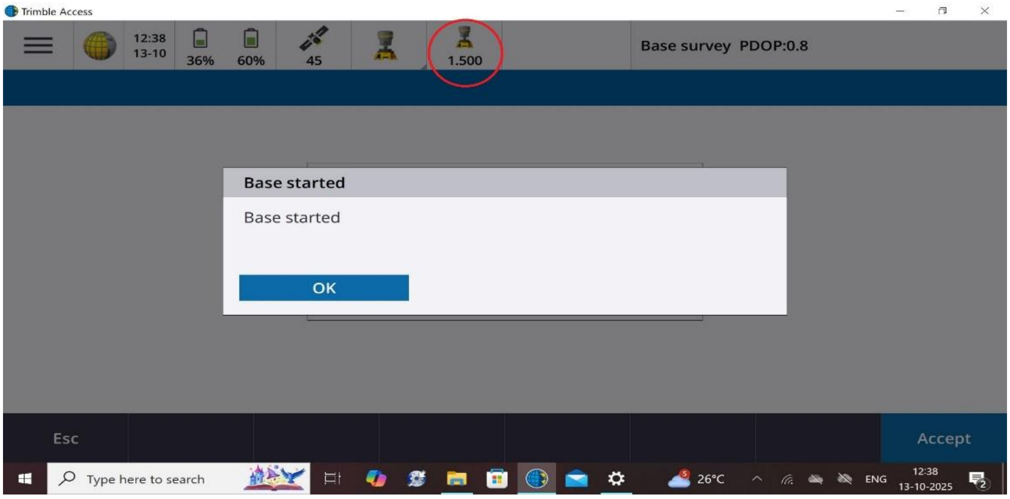


- Click >> **Store**

- Click >> **Start**

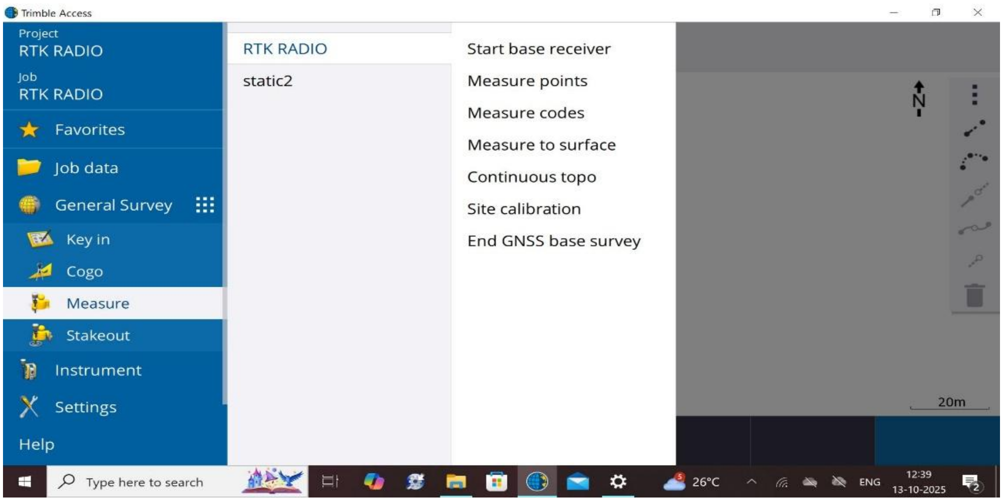


- Click >> **Ok** on **Base started**, here one can see there is a symbol of Base with it's height in red circle.

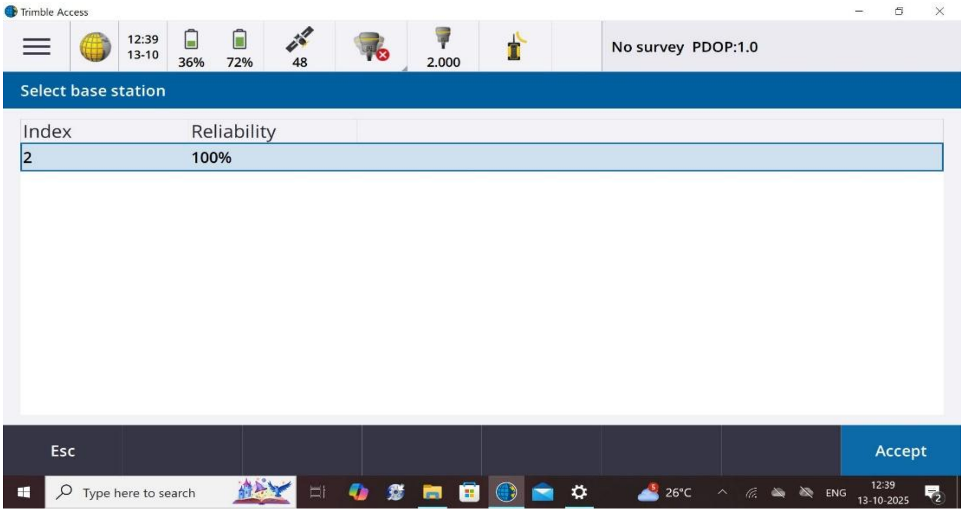


12.2 Rover initializing

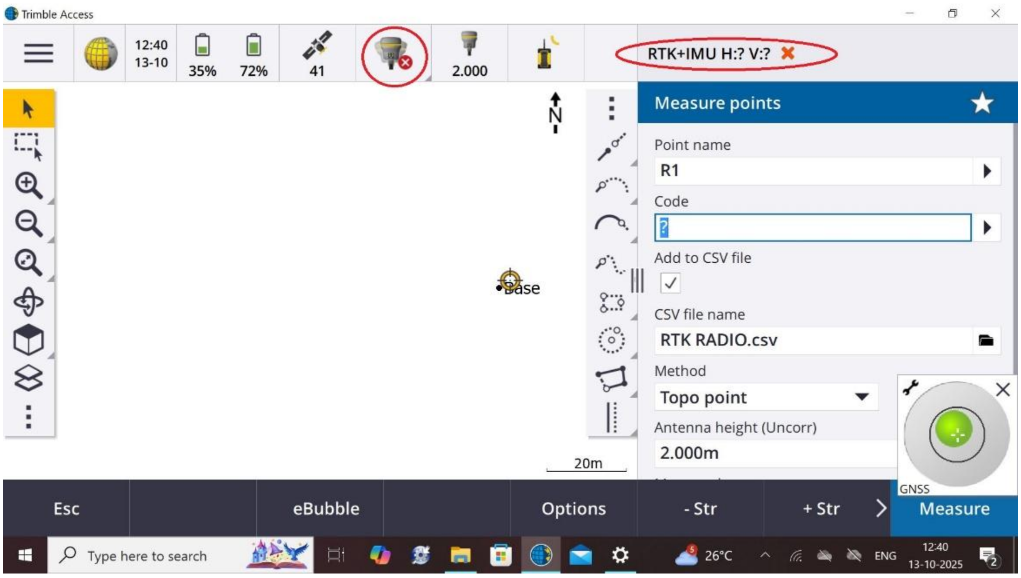
- Click on Menu >> **Measure** >> **RTK RADIO** (created survey style in earlier steps) >> **Measure points**



- Now **Select base station** index number here it is **2** (previously given), If more than one Base stations are being operated in working area then from this option they can be identified.

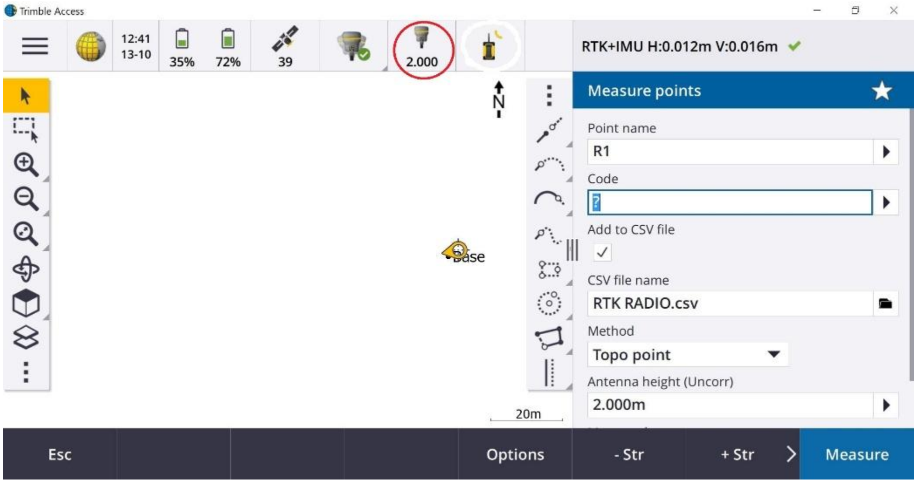


- ➢ Click >> **Accept**
  - Now in upcoming window provide **Point name**, **Code**, **CSV file name**, **Observing method** (Topo point, Rapid point or Observed Control point), recheck **Antenna height** & **Height measuring method** and check on **Add to CSV** . But here as Rover is not IMU calibrated so we can see Red cross marks on two places as marked in Red Circles. Also Click on **Options** icon and select **QC1&QC2** parameters and increase time of observation if required.

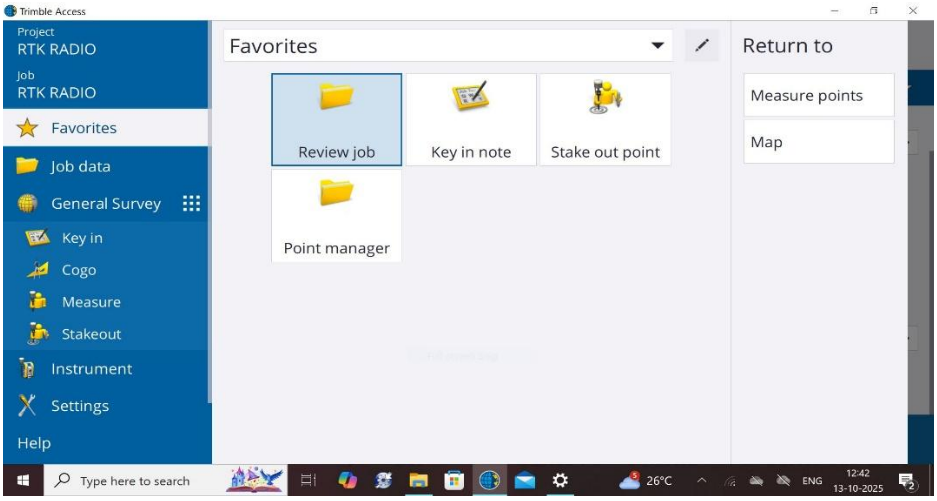


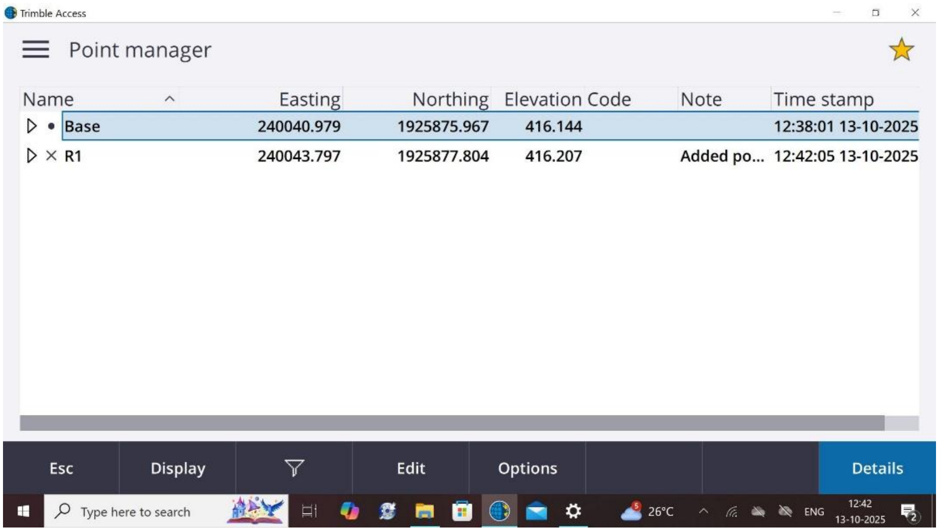
- ➢ In order to IMU calibrate Rover, Rover has to be moved in circles keeping it on Survey pole, also the tip of Survey pole is to be fixed on ground. Afterwards all Red cross signs would become Green Tick. Also we can see the Rover sign with height of Survey pole in Red circle.

- ➢ Now Click >> **Measure** >> **Store**



➢ Now to see observed points Click on Menu button, then Click >> **Favorites** >> **Point manager.** A list would be opened with all observed points.





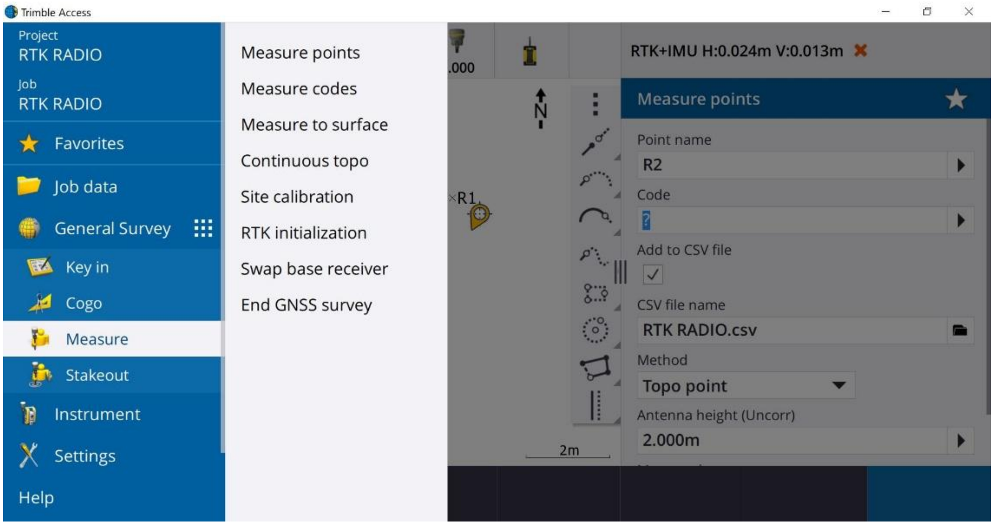
➢ Click >> **Esc.** Observations has been completed.

13. Closing observations of Rover and Base

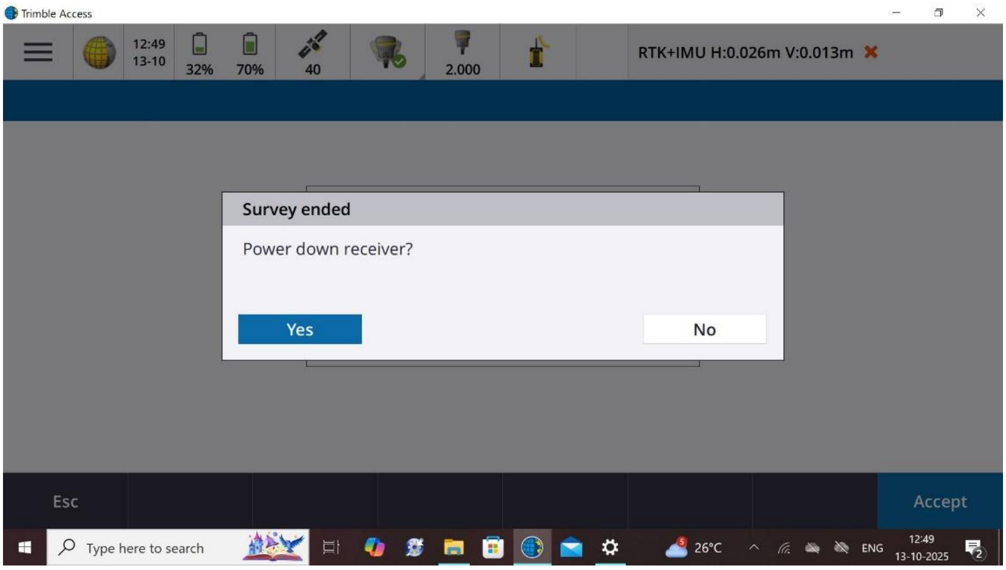
While closing RTK Observations first Rover should be shut down before turning off Base.

13.1 Closing of Rover

- ➢ Click on Menu Button then Click >> Measure >> End GNSS survey

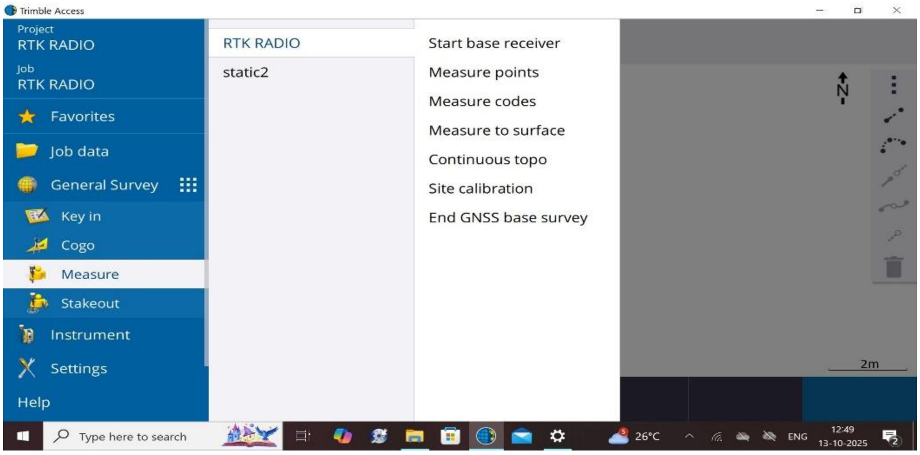


- ➢ To power down Receiver Click >> **Yes**

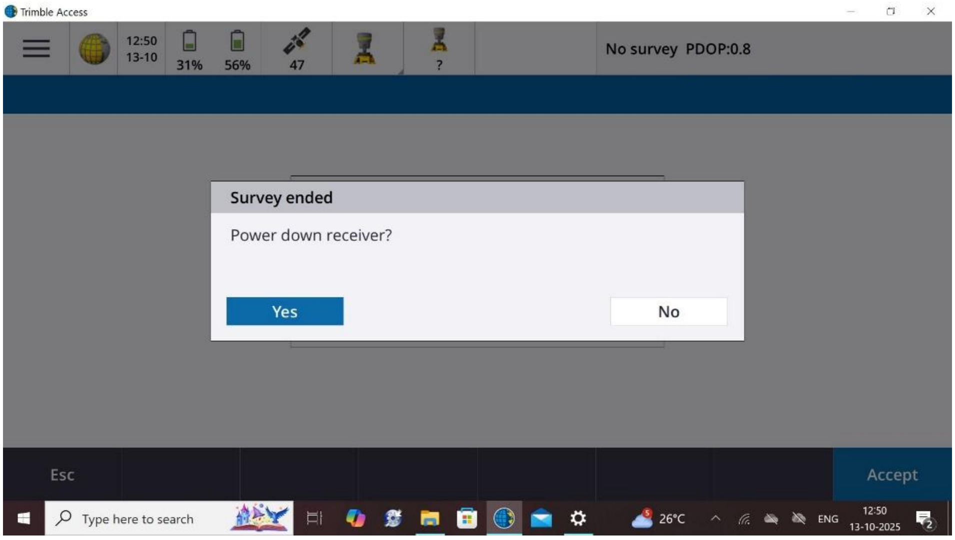


13.2 Closing of Base

- Click on Menu Button then Click >> Measure >> End GNSS base survey

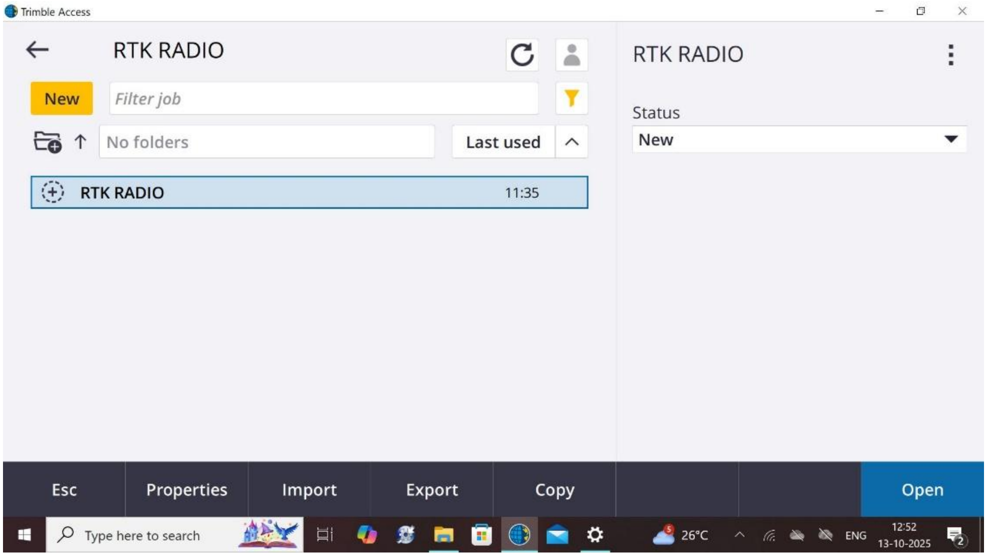


- To power down Base Click >> **Yes**

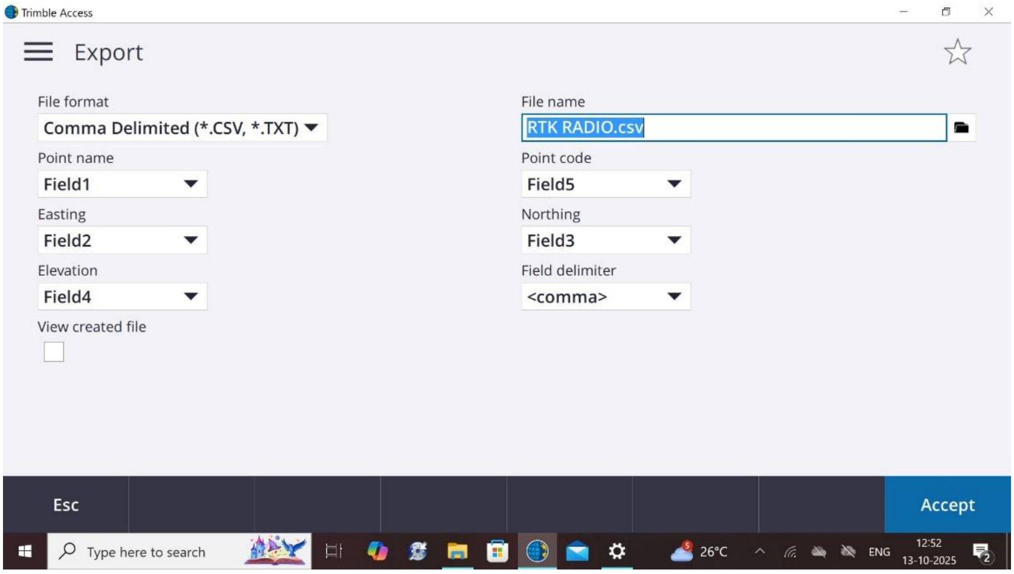


14. Downloading the Data

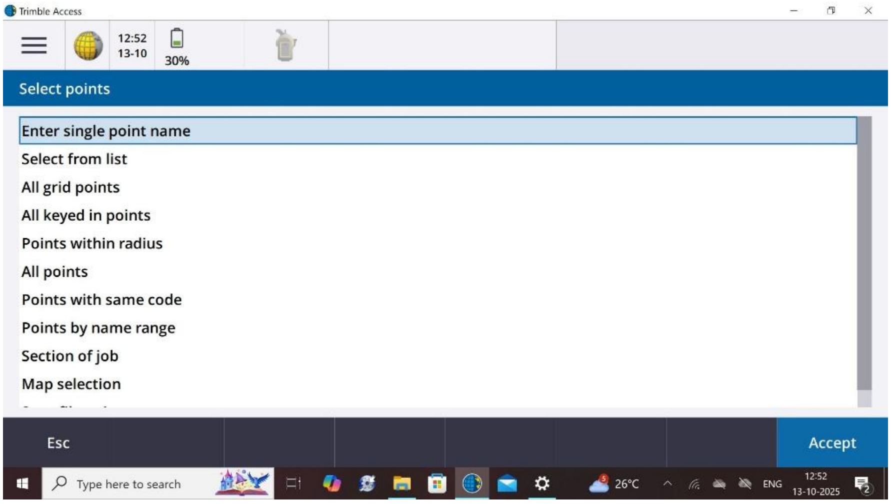
- After Ending the survey, Go to job data in menu of Trimble Access and Open the Project
  Select the Job **RTK RADIO** under Project **RTK RADIO**
  At the bottom you will find **Export** option >> Click

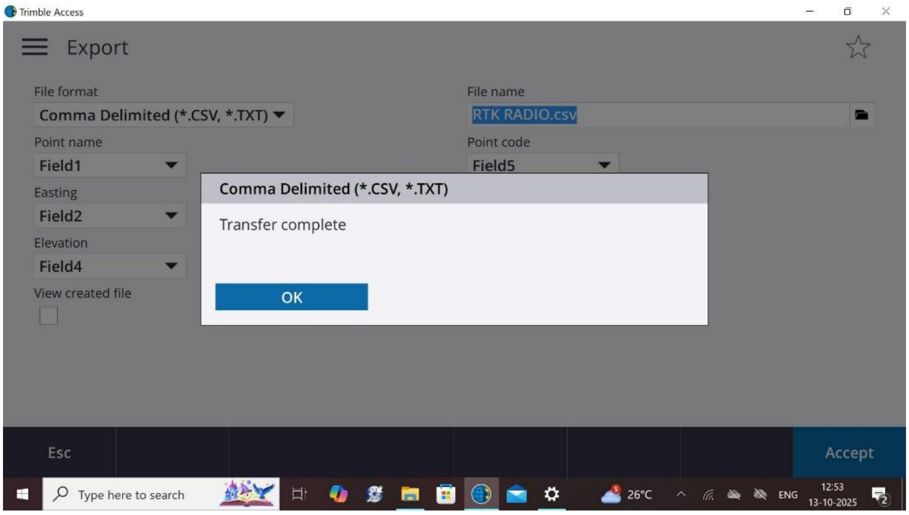


- Here in Export window you can select your desired format for observed data, also you can give path where exported data would be stored. Here we are selecting *.CSV format for our observed data. Click >> **Accept**



- Click on **All points**





➢ Click >> **OK**

15. Conclusion

This Trimble R980 RTK SOP confirms that all prescribed procedures for equipment setup, calibration, observation, and quality control were properly followed. By adhering to the SOP, reliable RTK fixed solutions with the required accuracy can be achieved, ensuring high-quality positional data. The standardized workflow enhanced field efficiency, minimized errors, and ensured consistency and compliance with survey accuracy requirements for precise GNSS data collection.

16. Annexure A: E-bubble calibration

E-bubble calibration

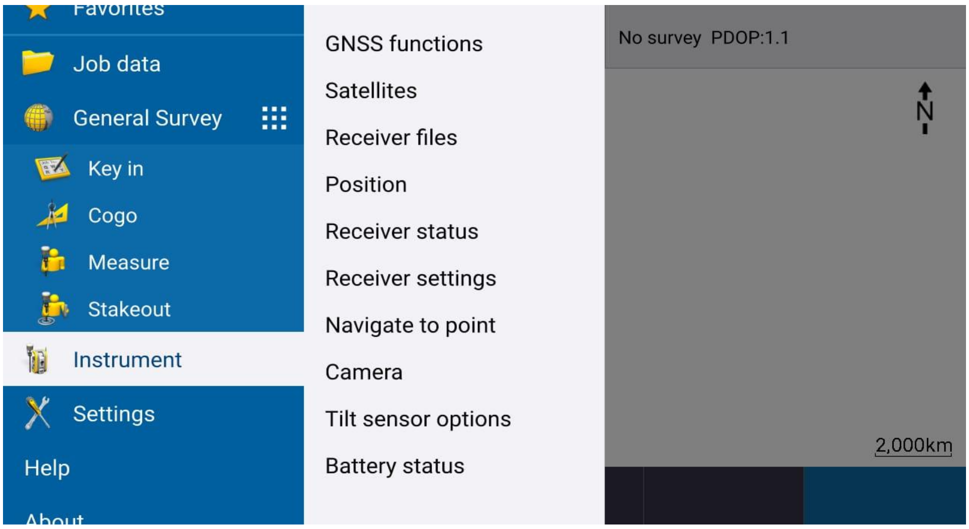
The following steps will guide you in performing the e- bubble calibration.

1. Set up the Receiver

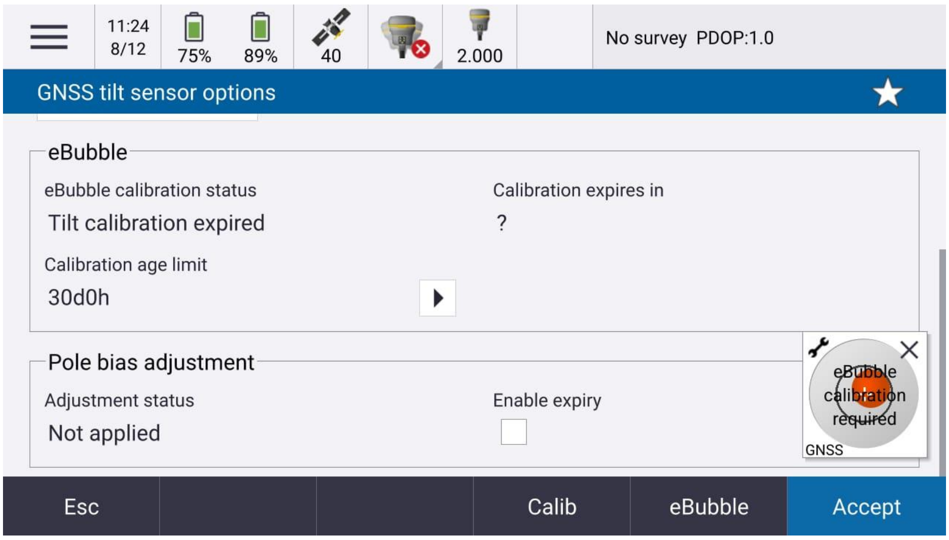
    - Connect R980 to controller via Bluetooth.
    - Mount the receiver on a bipod.
    - Level it using the bubble level.
    - Confirm stability before proceeding.

2. Make sure you properly levelled the receiver on bipod.

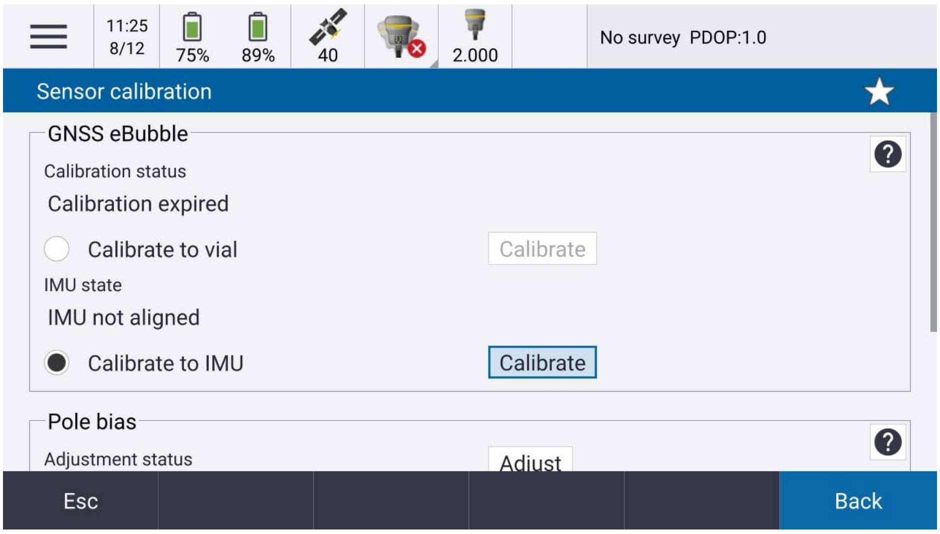
3. Open any created Job. Click on Menu Button >> **Instrument** >> **Tilt sensor options**



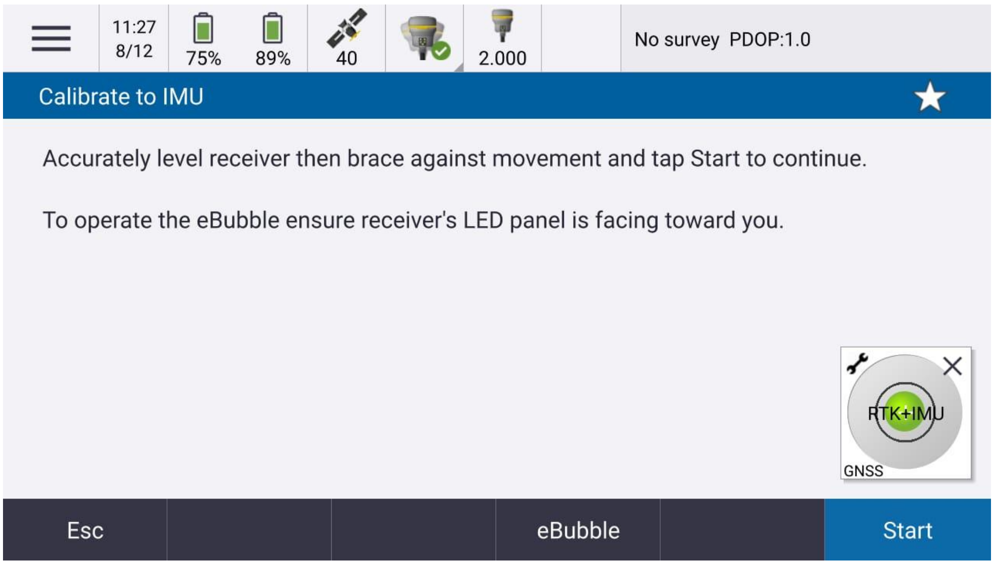
4. In **GNSS Tilt sensor** options Click on **Calib** button at bottom.



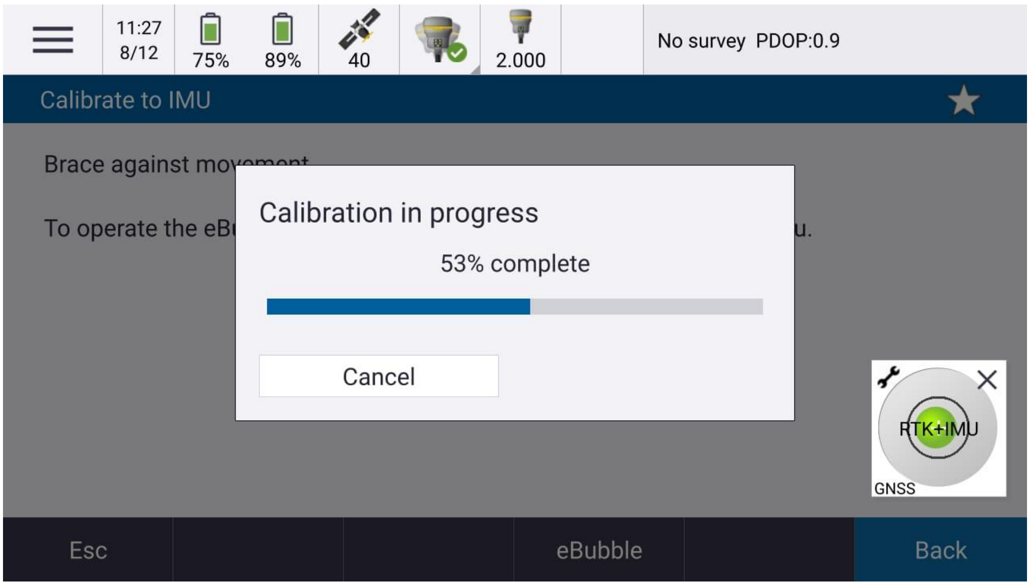
5. Now in upcoming window **Sensor calibration** select **Calibrate to IMU** and Click on **Calibrate** button.



6. Ensure the receiver is properly leveled and stand with controller facing the LED panel of the receiver and Click on **Start** button.



7. Next you can see the calibration in progress



8. After completion of calibration e- bubble level will match to the level on the receiver connected to the bipod and calibration process is completed.
9. E-bubble calibration has to be observed periodically and must be performed if required.

Note:- This maneuver should be done only when Controller is well connected to Receiver and observations are about to be start (RTK+IMU initialization).

IMU Alignment/Calibration

In order to IMU Alignment of Receiver, after complete setup of Receiver & Controller and just before the commencement of observations, Receiver mounted on Survey pole should be moved in round motion while keeping the tip of Survey pole tip stable on ground.

17. Annexure B : Various Positioning methods

Various Positioning methods in RTK/NRTK

In Trimble Access application following positioning methods are provided to **Measure Point**:-

- Topo Pont
- Rapid Point
- Multi Tilt Point
- Observed Control Point

Topo Point

A **Topo Point** is a point measured to **capture ground features** and terrain details during a **topographic survey**.

Rapid Point

A **Rapid Point** is a point observed with **short-duration repeated measurements** to obtain **better accuracy than a topo point**, but **without the full rigor of a control point**.It is commonly used in **RTK surveys** when you want **extra confidence** but **faster work**.

Multi Tilt Point

A **Multi-Tilt Point** is a point measured **several times at different pole tilt angles** using a **tilt-compensated GNSS receiver** (like Trimble R980), to **verify and improve confidence** in the measured position.

Observed Control Point

An **Observed Control Point** is a **high-accuracy reference point** whose coordinates are **established by observation** (GNSS static, fast static, or repeated RTK).

Comparison

Point Type	Method	Accuracy	Use
Topo Point	Single vertical	cm	Mapping Grade
Rapid Point	Repeated (time-based)	cm (better)	Temporary control
Multi-Tilt Point	Repeated (tilt-based)	cm (high confidence)	Validation / checks
Observed Control Point	Long static / repeat	mm–cm	Primary control (Survey Grade)

Continuous Topo

Continuous Topo is a feature-mapping method in Trimble Access that allows you to collect many topo points automatically while moving, instead of storing points one-by-one.

It continuously records positions at a set time or distance interval while the rover is moving with RTK fixed solution.

Ideal for linear features like Road, Railway, Tank Bund etc.

For getting observations in Trimble Access in ANY method whether it is RTK,NRTK OR PPP (RTX), Click on **Menu** Button >> **Measure** >> **RTK RADIO** (created survey style) >> **Continuous Topo .** In upcoming window we have to fill the starting **Point Id** and select **Method** (e.g. Fixed time etc. and enter that time interval in which you want the points) AND rest settings are same like in **Measure point** .
Now **Click** on **Start** tab in Right Corner in bottom, after this you will get points in a series within given time interval. To see these points you may go to **Point Manager**.

18. Annexure C : Stakeout

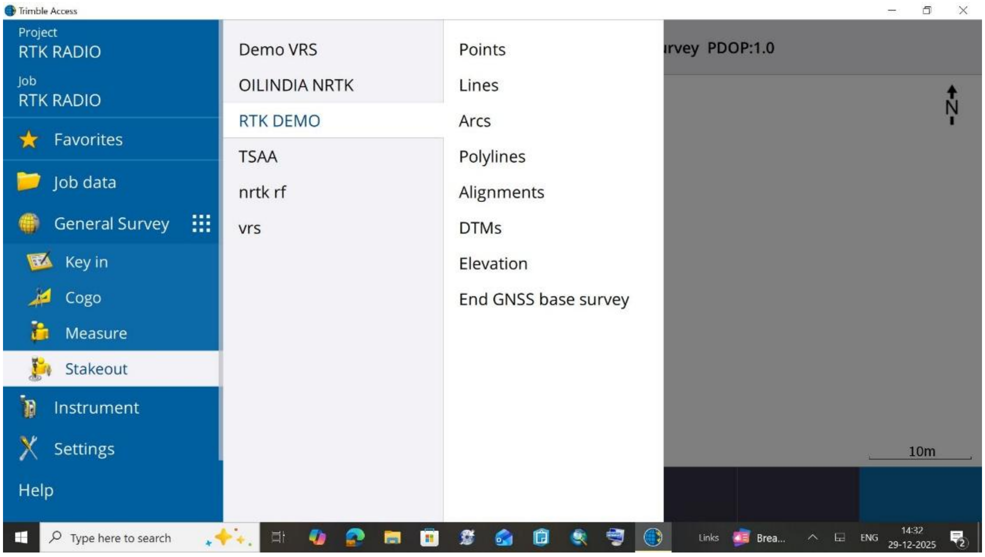
Stakeout

Stakeout is the process of **setting out (marking) design points on the ground** using GNSS guidance.

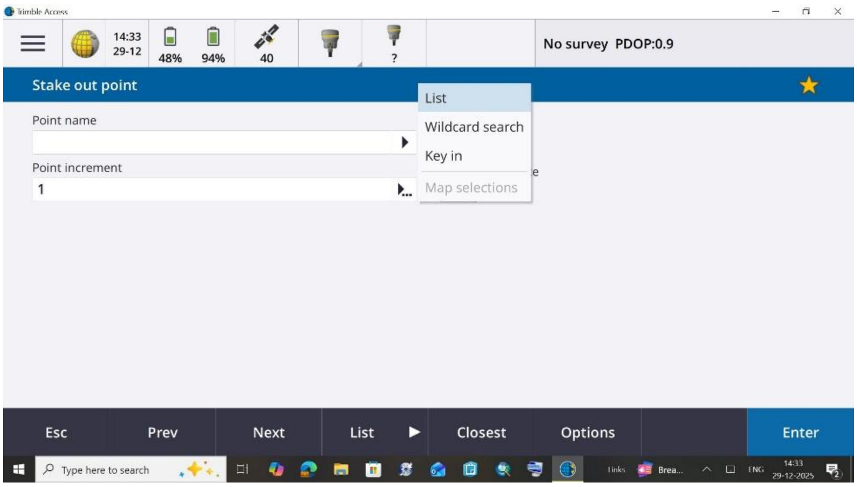
Purpose of Stakeout is to transfer design co-ordinates on ground. It is useful in Pillar/ Boundary points relocation.

Following is the process to do Stakeout in Trimble Access:

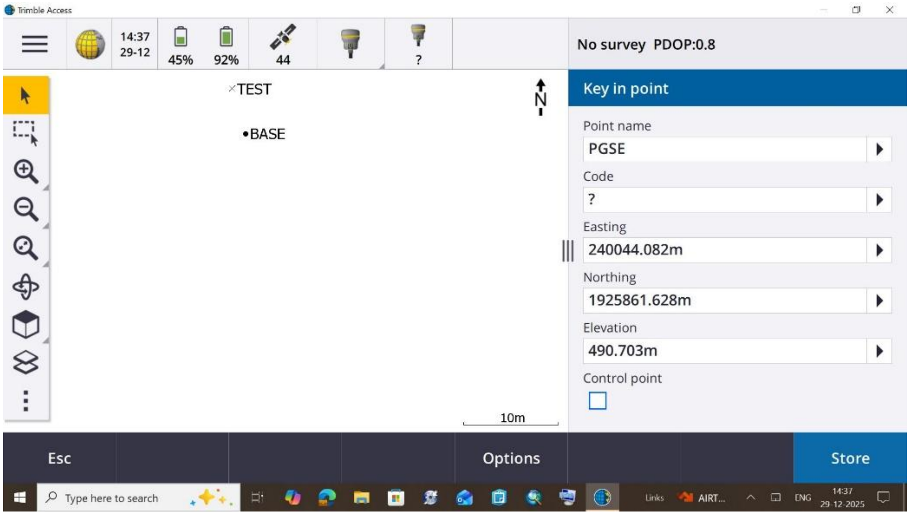
- Click >> **Menu** >> **Stakeout** >> **RTK DEMO** (Created style) >> **Points**

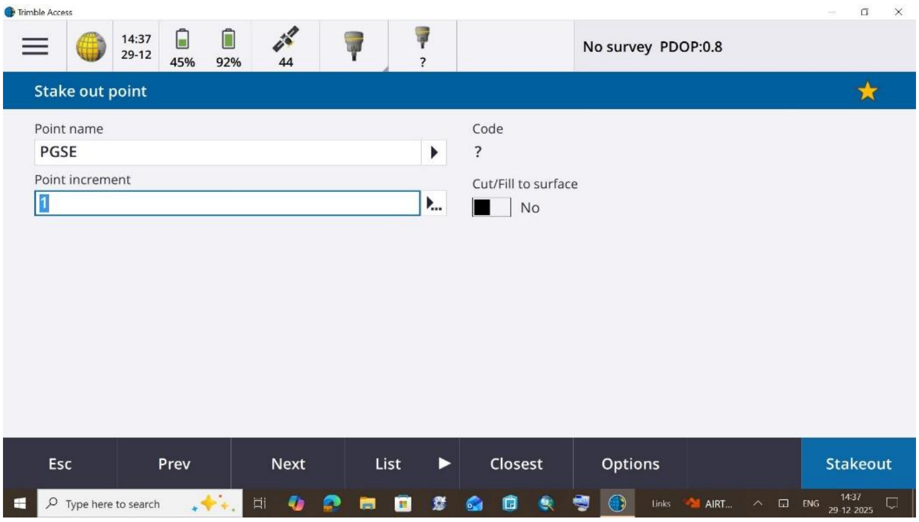


- In upcoming window, select Point to be relocate on ground, either by already installed **List** in the controller or **Key in** the co-ordinates of point to be relocate.

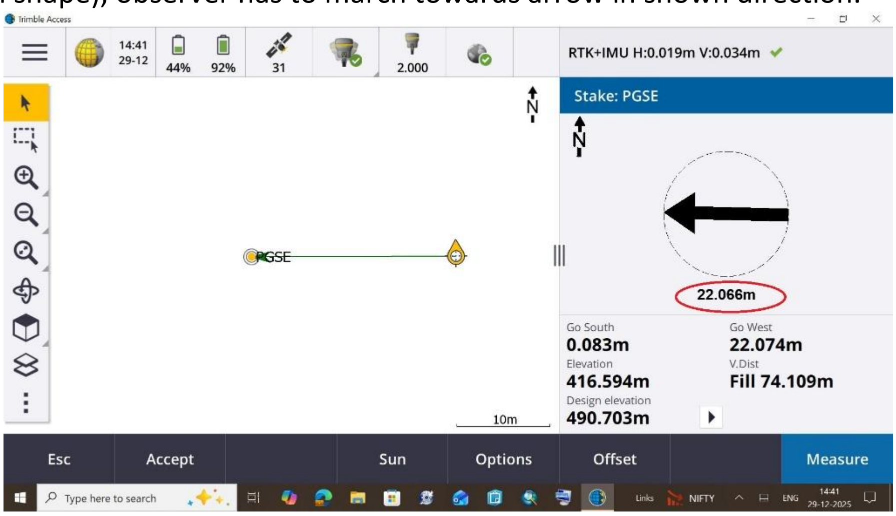


➢ Insert the co-ordinates Click >> **Store** >> **Stakeout**

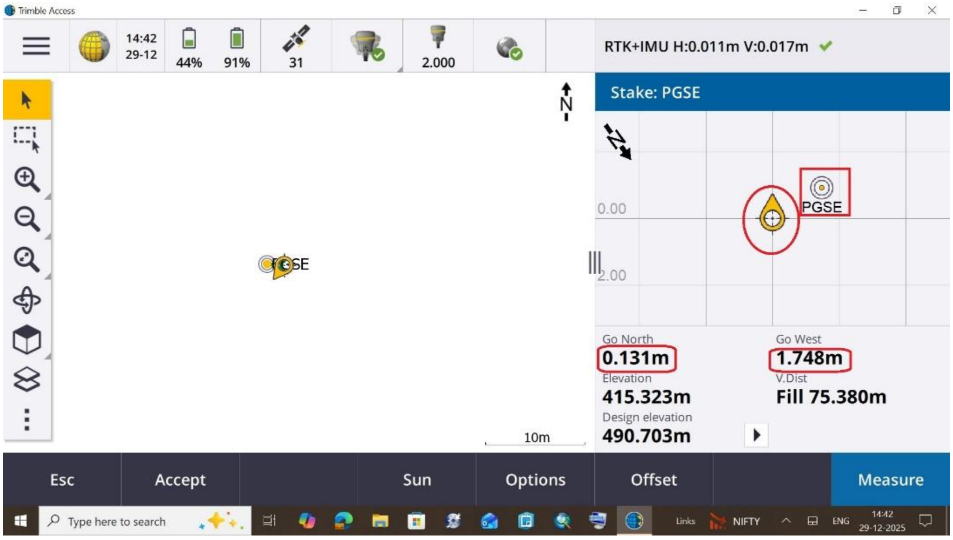




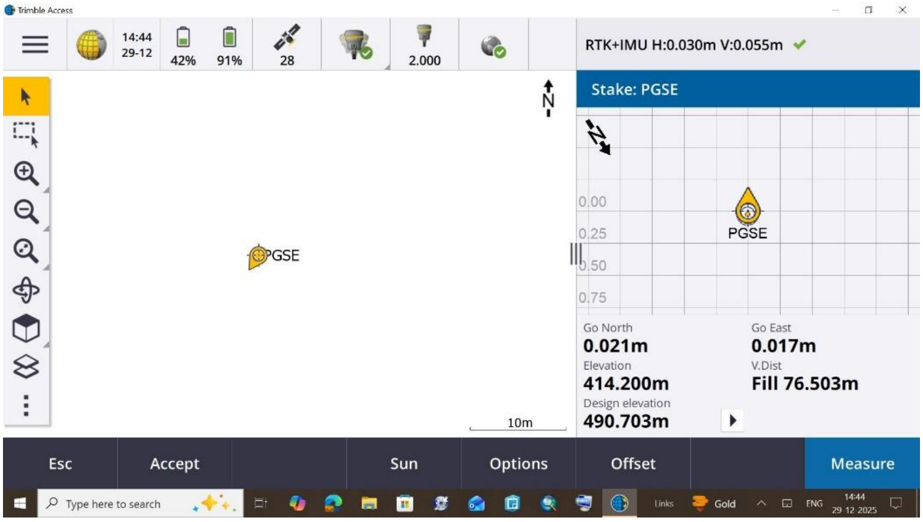
➢ In upcoming window distance between desired point & Rover is shown (in marked Red oval shape), observer has to march towards arrow in shown direction.



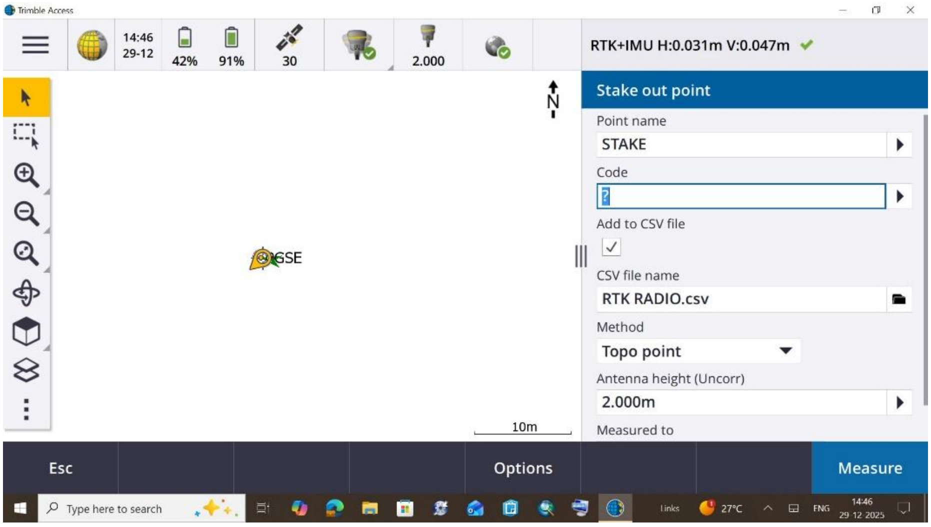
➢ When Rover comes near to desired point , a grid window appears wherein desired point (In marked Red color oval shape) and Rover (In marked square shape) are shown, also the distance in Easting and Northing also appears below the Grid box as follows:



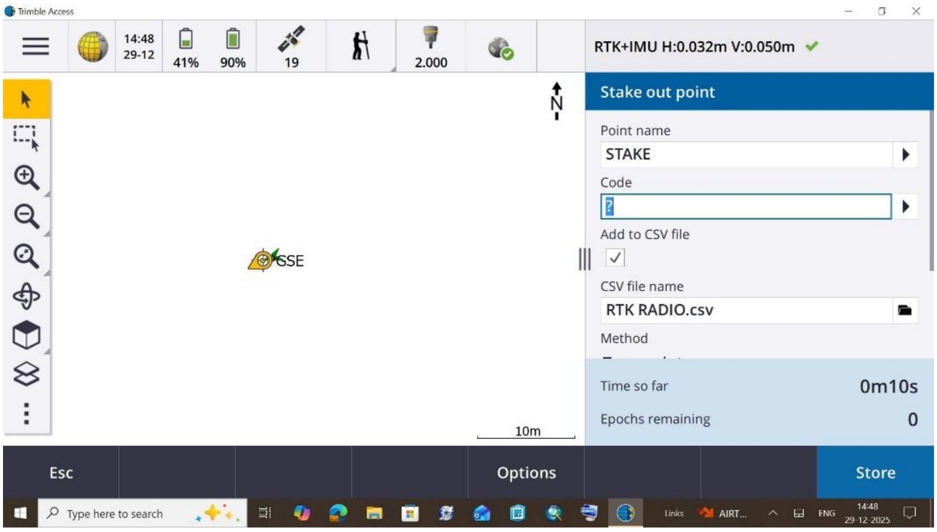
➢ When Desired position is coincided by Rover then Click >> **Measure**



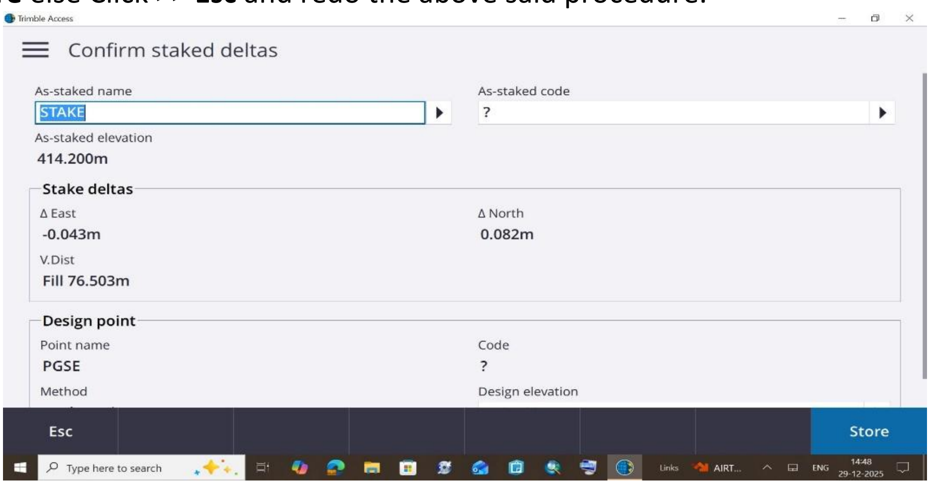
➢ In next window enter name of **Stake out point** name e.g. STAKE then Click >> **Measure**



➢ Click >> **Store**



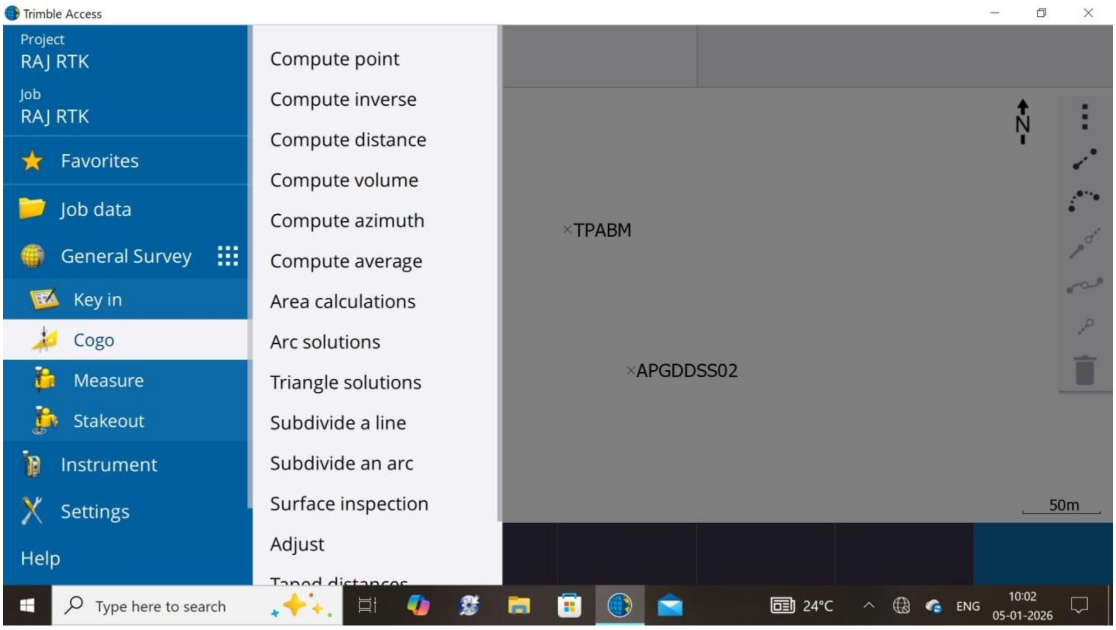
➢ In last step Stake out Deltas are shown, if they fulfill the stakeout criterion then Click >> **Store** else Click >> **Esc** and redo the above said procedure.



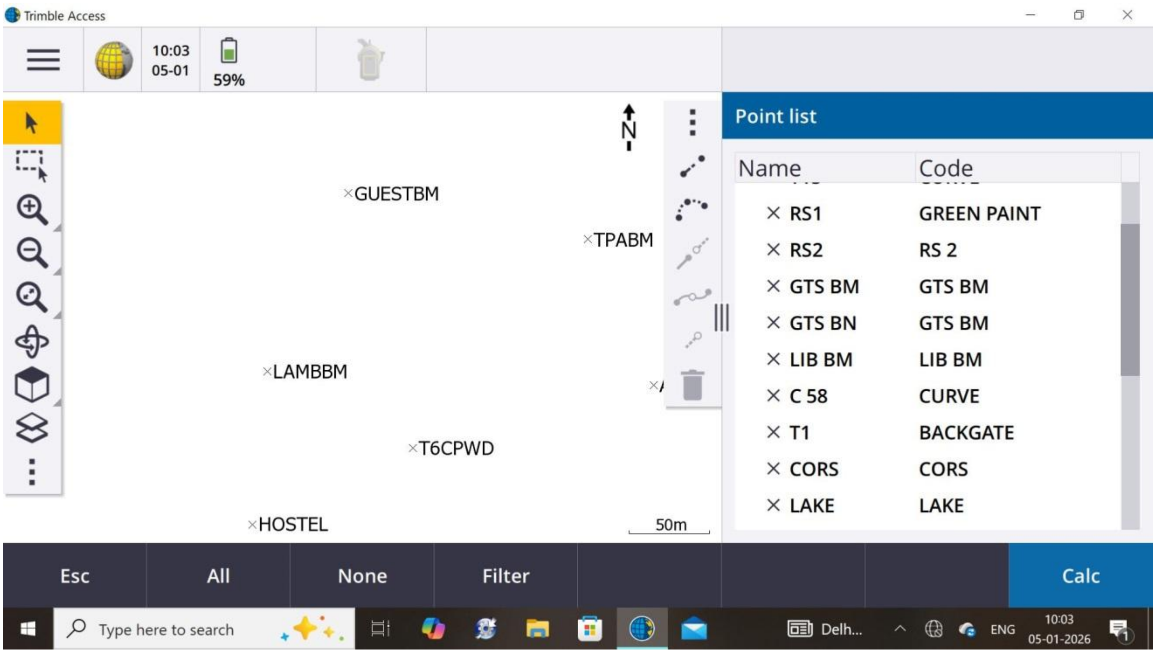
19. Annexure D : Area Calculation

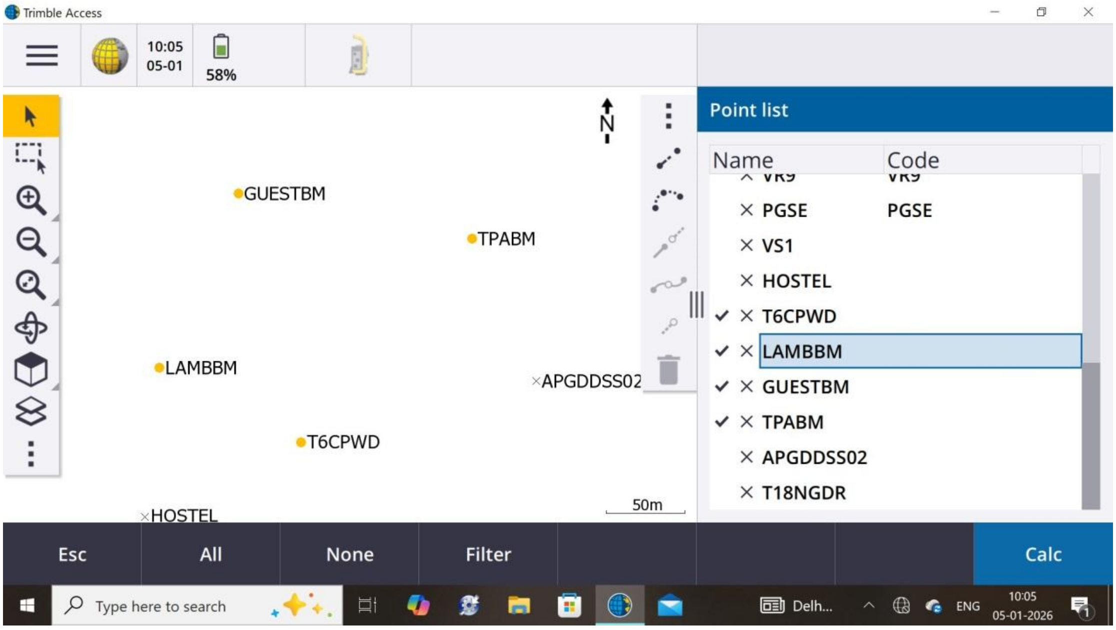
For area calculations using observed points (via NRTK/RTK/RTX) we have to follow below steps:-

- Open Project and Job then Click >> **Menu button** >> **Cogo** >> **Area calculations**

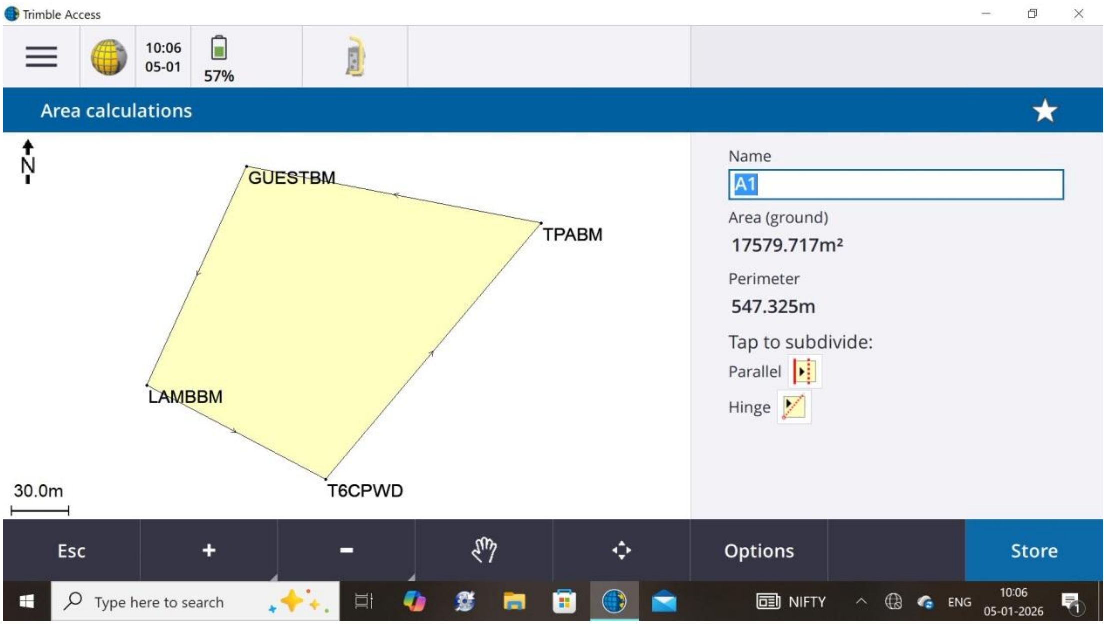


- A list of observed points will appear in the right side, we have to select all corner points of the desired area either in clockwise or in anti-clockwise manner.





➢ Click >> **Measure** and area would be shown like as below, we can save this by assigning a name.

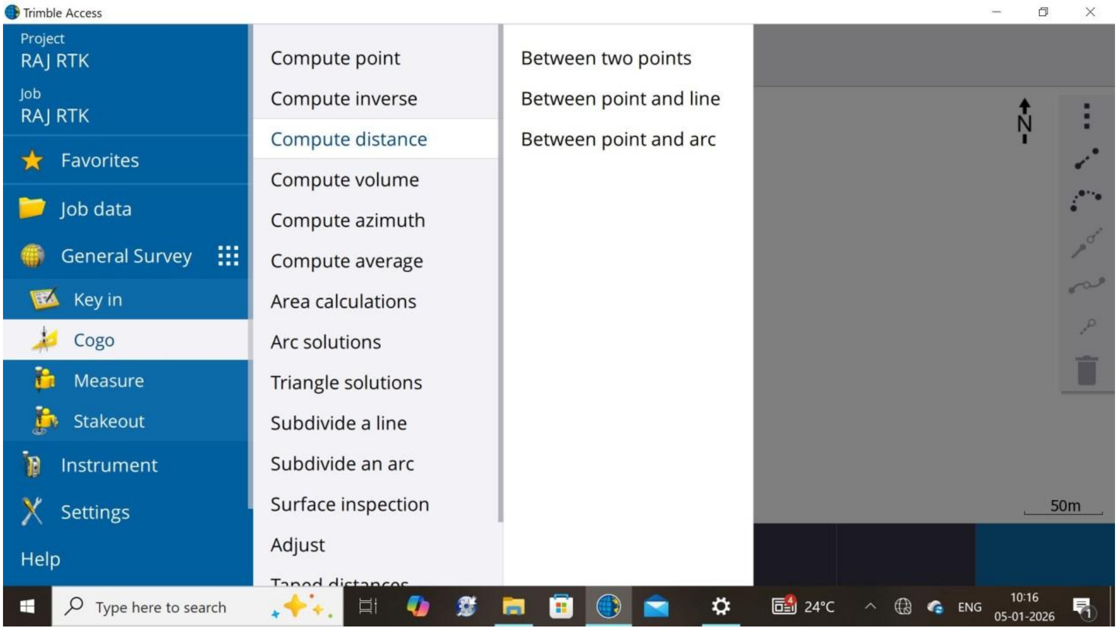


➢ Click >> **Store**

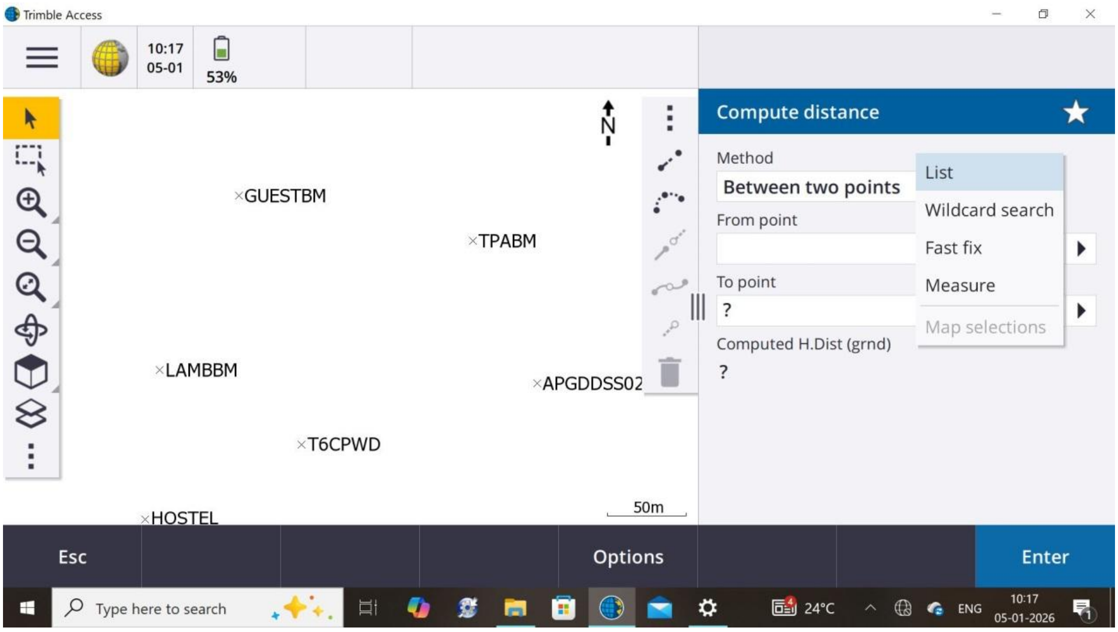
20. Annexure E : Distance Measurement

For Distance measurements using observed points (via NRTK/RTK/RTX) we have to follow below steps:-

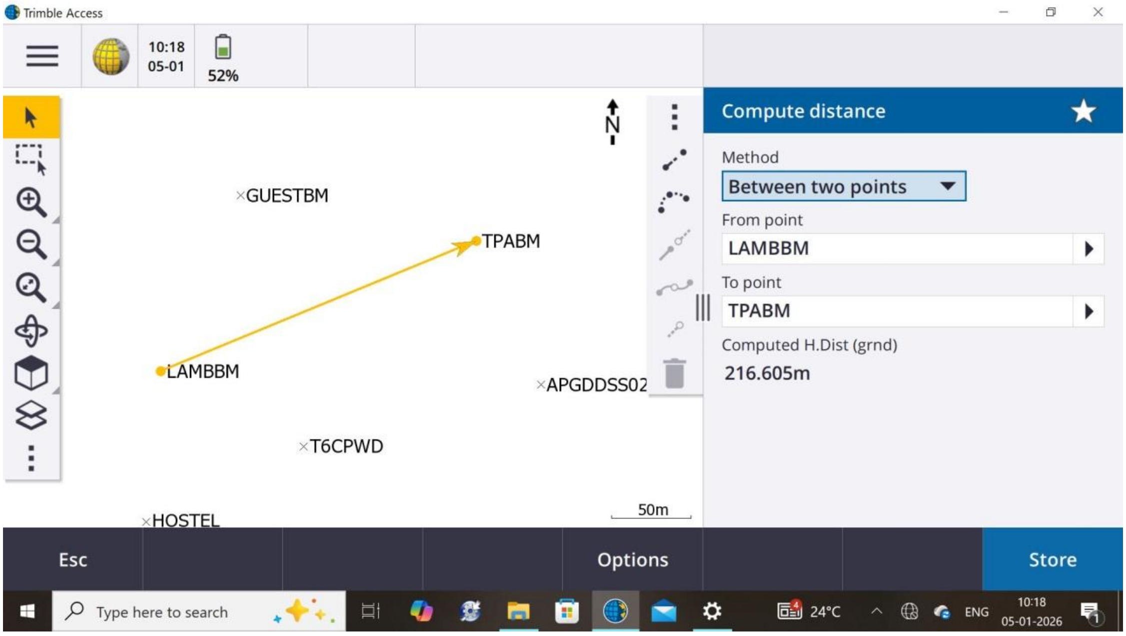
- Open Project and Job then Click >> **Menu button** >> **Cogo** >> **Compute distance >> Between two points/Between point and line**



- A box will appear in the right side as follows:



- Select method of computing distance e.g. Between two points and afterwards select two desired points



- Click >> **Store**